

The Blockwise Alperin Weight Conjecture

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Abstract

We prove the numerical blockwise Alperin weight conjecture for every finite group and every prime p . For each p -block B , the number of irreducible Brauer characters in B equals the number of conjugacy classes of B -weights.

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1 Introduction

The local–global principle in modular representation theory seeks to recover information about representations of a finite group from normalizers of its p -subgroups. Alperin’s weight conjecture predicts an exact numerical form of this principle, block by block: simple modules are counted by defect-zero ordinary characters of normalizer quotients. Alperin formulated the conjecture at the 1986 Arcata conference; the original account appeared in the proceedings in 1987 [1, 13].

Let p be a prime and G a finite group. For an algebraically closed field k of characteristic p , a block is a primitive central idempotent of kG . For a block B , let $l(B)$ be the number of simple kG -modules belonging to B , equivalently the number of its irreducible Brauer characters. For character calculations we use a splitting p -modular system $(K, \mathcal{O}, k)$: $\mathcal{O}$ is a complete discrete valuation ring with characteristic-zero fraction field K and residue field k , and K splits every subgroup of G . A block also denotes its unique central lift to $\mathcal{O}G$. In Section 2.1 we justify reduction to $k = \overline{\mathbb{F}}_p$ and construct such a system; the resulting counts and block assignments are unchanged over any algebraically closed extension of this field.

For a positive integer a , write a_p for its largest power-of- p divisor. A p -weight of G is a pair (Q, ϕ) , where $Q \leq G$ is a possibly trivial p -subgroup and

$$\phi \in \text{Irr}(N_G(Q)/Q), \quad \phi(1)_p = |N_G(Q)/Q|_p.$$

Thus ϕ is an ordinary irreducible character of defect zero in the quotient. The block assigned to this weight is defined using the block b of the inflated character in $H = N_G(Q)$.

We specify the block-assignment convention. For $H \leq G$, coefficient restriction is the linear map

$$s_H : Z(kG) \longrightarrow Z(kH), \quad \sum_{g \in G} a_g g \longmapsto \sum_{h \in H} a_h h.$$

If c is a block of a finite group, its central character λ_c is the residue character of the corresponding local factor of the center. In particular it is the scalar action of the center on any simple module in c . We say that $b^G = B$ when

$$\lambda_b \circ s_H = \lambda_B. \tag{1.1}$$

This is the central-character definition of Brauer block induction [6, p. 338]; see also the corrigendum [7]. In the normalizer situations arising from weights, the composite is indeed a central character; we prove this directly in Section 3. Let $W_p(B)$ be the set of simultaneous G -conjugacy classes of weights assigned to B .

Theorem 1.1. *For every prime p , every finite group G , and every p -block B of G ,*

$$l(B) = |W_p(B)|.$$

Theorem 1.1 resolves the numerical blockwise Alperin weight conjecture positively, with no restriction on the defect groups, the prime, or the finite group. The case $Q = 1$ is part of the assertion. The conclusion is an equality of cardinalities; a canonical or equivariant correspondence is not required for this numerical statement.

Local–global consequences

The numerical equality also gives the following established consequences of the weight conjecture. Write $\text{IBr}_p(H)$ for the set of irreducible p -Brauer characters of a finite group H .

Corollary 1.2 (Local–global and principal-block bounds). *Let p be a prime, G a finite group, B a p -block with defect group D , and b its Brauer correspondent in $N_G(D)$. Then*

$$l(B) \geq l(b), \quad D \text{ abelian} \implies l(B) = l(b).$$

For $P \in \text{Syl}_p(G)$, one also has

$$|\text{IBr}_p(G)| \geq |\text{IBr}_p(N_G(P))|.$$

Let B_0 be the principal p -block of G , and let $k_p(G)$ count the conjugacy classes of p -power-order elements, including the identity. Then

$$\begin{aligned} k_p(G) = 3 &\implies l(B_0) \geq (p-1)/2, \\ p \mid |G| &\implies l(B_0) \geq 2\sqrt{p-1} + 1 - k_p(G). \end{aligned}$$

The local block bounds are Alperin’s Consequences 1–2 [1]; the principal-block bounds are due to Hung, Sambale, and Tiep [22, Propositions 3.1–3.2]. Their weight-conjecture hypotheses are supplied by Theorem 1.1. The complete derivation, including the passage to the block-free inequality, appears in Section 8.

The block-free inequality for $p = 2$ was already proved by Malle–Navarro–Tiep [28, Theorem B]; the local inequality for maximal-defect 2-blocks was proved by Feng–Martínez–Rossi [15, Theorem D].

The sharper classification of character-count and defect data, and the functorial formulation of the theorem, are given in Section 8. We first explain the historical setting and the proof of the local–global equality itself.

Prior work and the proof’s ingredients

Alperin’s formulation extends the notion of weights from the modular representation theory of finite groups of Lie type in defining characteristic to arbitrary finite groups [2]. Early large-family cases include Alperin and Fong’s work on symmetric groups and on finite general linear groups in odd modular characteristic [2]. Knörr and Robinson expressed the conjecture as alternating sums over chains of p -subgroups [25]. Navarro and Tiep reduced the non-blockwise conjecture to conditions on finite simple groups [31], and Späth obtained a blockwise reduction [35]. These reductions organize the problem around local correspondences for simple groups. Among the resulting advances, Feng, Li, and Zhang proved the blockwise conjecture for finite special linear and special unitary groups and for finite groups with abelian Sylow 3-subgroups [13]. Feng, Malle, and Zhang subsequently studied generic weights for finite reductive groups under explicit hypotheses [14].

Other recent progress includes the inductive blockwise condition for $F_4(q)$ at odd primes not dividing q , proved by An, Hiss, and Lübeck [3], and Hu and Zhou’s results on inertial blocks, including an alternative proof for 2-blocks with abelian defect [20]. Huang and Yılmaz develop chain, Galois, and functorial reformulations [21]. A recent preprint of Zhang claims the inductive condition for types B, C and sporadic groups [39]; those classes do not exhaust the scope of Theorem 1.1. Our proof instead passes from a direct block-algebra calculation to covers of curves; it does not use a classification of finite simple groups or an inductive weight condition.

The algebraic part uses classical constructions with a specific counting purpose. The subgroup-chain argument is related to the Knörr–Robinson reformulation. Powers modulo commutators belong to Külshammer’s theory of generalized Reynolds ideals [18], and conjugation averaging is the group-algebra Higman trace [19, 27]. The sum of the p -elements detects defect-zero blocks as in Tsushima’s work [38]. The passage from central traces to tuples is a weighted genus-one form of Frobenius’s character-counting formula; see [24] for the surface-group formulation. We prove the

exact versions needed here, including the block projections and normalization of the tuple count. The further assertion is uniform divisibility as the number of punctures increases, not merely a character formula for a fixed surface.

Chen's relation between Nielsen orbit counts and degrees of moduli of elliptic covers [8] informed the geometric divisibility viewpoint. The many-marked, high-index construction and the uniform puncture-divisibility theorem required here are developed in this paper, not supplied by Chen's theorem.

The geometry combines several established ideas with that divisibility problem. High-index genus-one curves are classical objects in period-index theory; Clark and Lacy prove arbitrary-index existence over infinite fields finitely generated over their prime fields [9]. Our construction also prescribes marked points giving independent absolute field differentials. The genus calculation belongs to the theory of inseparable genus change initiated by Tate [36]; see Schröer [33] for a modern treatment. Its height-one determinant identity is a curve-level form of the canonical-bundle formula for p -closed foliations [12]. Here the calculation must keep additional constants and inseparable residue degrees, and the large index turns a bounded-error estimate into an exact inequality. Finally, maximal-immediate-extension methods of Krull and Kaplansky [26, 23, 32] provide a suitable valued field, while the center-valued obstruction studied by Dèbes and Douai [10] explains the arithmetic descent issue for marked covers. We give the required constructions and descent argument directly.

The mechanism of the proof

Chain inversion first reduces the weight equality to an identity between a defect-zero character count and an alternating sum of simple-module counts in subgroup normalizers. A calculation with powers modulo commutators, followed by cancellation of chains, replaces those simple counts by ordinary-character counts. We then construct a central group-algebra element supported on p -singular elements, with residue scalars zero or one chosen to remove the defect-zero contribution from the alternating sum. Lift it by class sums to characteristic zero and restrict it to each chain normalizer. The alternating sum of traces of the p^s th powers converges to the difference between the alternating ordinary-character sum and the defect-zero count. Expanding the traces gives weighted tuple counts. A further chain cancellation leaves only tuples generating subgroups with no nontrivial normal p -subgroup. The complete reduction is [Proposition 3.14](#).

Here is the counting problem that remains. Fix a finite group J with $O_p(J) = 1$, where $O_p(J)$ is its largest normal p -subgroup, and set $n = p^s$. We count ordered tuples $(x, y, u_1, \dots, u_n) \in J^{n+2}$ satisfying

$$[x, y]u_1 \cdots u_n = 1, \quad \langle x, y, u_1, \dots, u_n \rangle = J.$$

Here $[x, y] = xyx^{-1}y^{-1}$, and each puncture entry u_i is p -singular: its order is divisible by p . Prescribe a multiset of n conjugacy classes for the u_i , and identify tuples only by simultaneous inner conjugation in J , not by permutations of their positions. For every fixed integer $h \geq 1$, we must prove that all these counts are divisible by p^h for sufficiently large s ([Theorem 3.1](#)). The threshold may depend on p, J, h , but not on the multiset. This uniformity makes each weighted trace sum tend to zero even though the number of multisets grows with s .

The geometric construction supplies two properties on the same marked genus-one curve E/F_0 in characteristic p . First, every divisor degree on E is divisible by p^s . Second, after a cyclic splitting extension it becomes a constant elliptic curve with independently varying marked points. More precisely, put $\kappa = \overline{\mathbb{F}}_p$, choose an ordinary elliptic curve A_0/κ , and take $F' = \kappa(A_0^n)$. The extension F'/F_0 is cyclic Galois of degree n and splits E . Over F' , the marked points are the generic projection points $t_i \in A_0(F')$. If η is a nonzero invariant differential on A_0 , then the pullbacks $t_i^* \eta$ form a basis of $\Omega_{F'/\kappa}$. Their independence is in this field-differential space, not in the one-dimensional space of

regular relative differentials on the elliptic curve. After a finite extension F/F_0 with $[F : F_0]_p \leq p^h$, every divisor degree on E_F is still divisible by p^{s-h} , and at most h independent differential directions are lost over the compositum FF' (Proposition 4.4, Corollary 4.5, and Lemma 4.6).

For a purely inseparable cover of this curve, a calculation along successive maps of exponent one gives a genus bound with an error controlled by h and the cover degree. The index property makes this bound exact: after multiplication by the cover degree, the difference between the genus term and the ramification expression is a multiple of p^{s-h} . The differential calculation bounds that difference below by a negative constant independent of s . Once p^{s-h} exceeds the constant's magnitude, the difference must be nonnegative. This is the rounding step in Theorem 5.1. The resulting estimate is useful independently of the block-algebra reduction: it applies to regular curves over imperfect fields, retains inseparable residue degrees and extra constants, and does not require geometric reducedness.

We lift the marked genus-one curve to mixed characteristic and pass to a spherically complete valued field K_s with value group $\mathbb{Q}$ and residue field F_0 . Over an algebraic closure, the generating tuples are precisely the monodromy data of connected J -covers of the punctured genus-one curve, with the J -action specified. Simultaneous inner conjugacy changes only the chosen fiber point; it does not forget the ordering of the punctures. The lift contains all $|J|$ th roots of unity. Galois therefore preserves the local conjugacy classes as it permutes the marked points, so it acts on the finite cover set with the prescribed multiset. This action factors through a finite Galois group Γ ; take a Sylow p -subgroup of Γ .

An orbit of size less than p^h would give a cover defined over an extension L/K_s with $[L : K_s]_p < p^h$. The point requiring proof is that a fixed isomorphism class actually descends with its specified J -action. The obstruction is an extension with kernel $Z(J)$. Since $O_p(J) = 1$, this kernel has order prime to p , and an elementary Sylow argument supplies a descent datum (Proposition 6.6). Put $F = \text{res}(L)$. The full degree of L/K_s equals $[F : F_0]$, so this residue extension also has small p -part.

To rule out such a descended cover Y , extend the valuation obtained by reducing functions on the base curve to the function field of Y , and take the finite J -orbit of one such extension. For each valuation w in this orbit, normalizing E_F in its residue field gives a regular projective curve C_w . Reduction of section spaces first shows that the degrees of the maps $C_w \rightarrow E_F$ sum to $|J|$. A second comparison imposes vanishing above selected marked points. Together with characteristic-zero Riemann–Hurwitz and the sharp inseparable genus estimate, these section comparisons force

$$\chi(Y) = \sum_w \chi(C_w),$$

where each structure-sheaf Euler characteristic is computed over its stated ground field. The equality gives enough sections to construct a finitely presented flat model whose special fiber is the disjoint union of the C_w (Proposition 7.5). Passing to a Noetherian normal base allows us to apply connectedness. There is therefore only one C_w , and its valuation is stabilized by all of J . The kernel of the action on its residue field, the inertia group, is consequently normal in J . The same genus comparison shows that this kernel is a nontrivial p -group, contradicting $O_p(J) = 1$.

Every Sylow orbit thus has size at least p^h . Its size is a power of p , hence is divisible by p^h . Summing the orbits proves the uniform tuple divisibility, and Proposition 3.14 returns this conclusion to the block identity.

The auxiliary fields in the geometric construction are unrelated to the coefficient system $(K, \mathcal{O}, k)$ used in the block count.

Figure 1 displays how the two geometric inputs exclude covers over fields whose degree has small p -part, and how this returns to the original block count. Section 2 records the classical inputs and coefficient choices. Section 3 proves the algebra-to-counting reduction before any curve is constructed.

Section 4 supplies the marked high-index curve, Section 5 establishes the sharp genus estimate, and Section 6 constructs the valued lift and descends small orbits. Section 7 builds the finite model, applies connectedness, and completes both the divisibility theorem and the block identity. Finally, Section 8 proves the announced local–global bounds and presents the character-count classification and functorial identity with their full conventions.

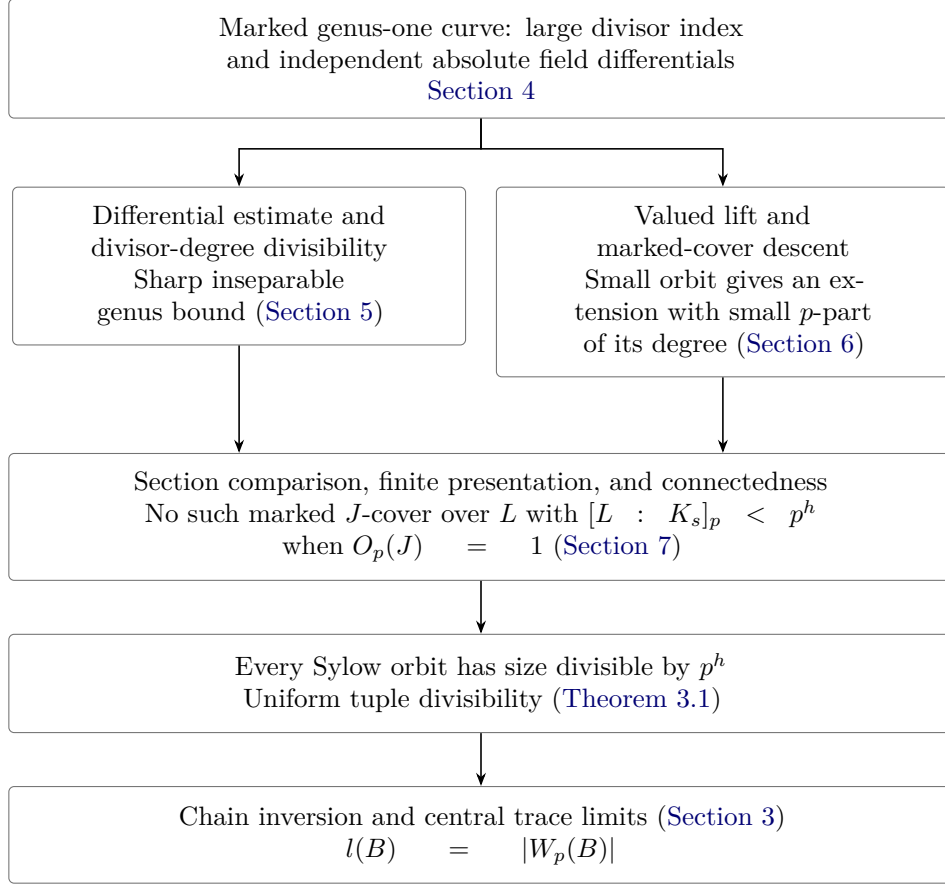


Figure 1: Mathematical dependencies, not reading order. The covers are geometrically connected J -Galois covers, with specified J -action over their field of definition, branched only at the marked points and with inertia order divisible by p at every marked point. The two branches provide, respectively, the genus estimate and the arithmetic realization of a hypothetical small Sylow orbit. For sufficiently large s , their combination rules out such a cover over a field whose degree has small p -part; the geometric cover degree is fixed at $|J|$. The resulting uniform divisibility supplies the counting hypothesis of the block-algebra reduction, which is proved before the geometric construction and applied at the end. No weight-counting theorem is used to construct the curves.

Conventions

All groups are finite. The identity is a p -element; a p -singular element has order divisible by p . We use $[x, y] = xyx^{-1}y^{-1}$ and write $O_p(X)$ for the largest normal p -subgroup of X . All curve degrees and coherent Euler characteristics are taken over the ground field specified at that point. A regular curve over an imperfect field is not presumed smooth or geometrically reduced. For such a projective

curve C , we set

$$\chi(C) = \chi(\mathcal{O}_C), \quad \nu(C) = -2\chi(C).$$

These quantities add on disjoint unions. A valuation is written additively. A field extension is *immediate* if it changes neither the value group nor the residue field.

2 Classical foundations used

We state the classical inputs to make the scope of the proof explicit. We also use elementary linear algebra, field and group theory, and the basic definitions and functorial formalism of schemes, sheaves, divisors, and Kähler differentials. The counting, index, inseparable-genus, valued-field, and specialization assertions particular to this proof are established in the later sections. In particular, no weight-counting theorem or conjectural reduction is an input.

Foundation 1 (Finite groups). For a finite group X , its p -subgroups lie in Sylow subgroups of order $|X|_p$. A nontrivial finite p -group has nontrivial center and an element of order p , and each of its maximal proper subgroups is normal. We use the orbit–stabilizer formula and the usual fixed-point and conjugation actions.

Foundation 2 (Finite-dimensional algebras and characters). The Jacobson radical of a finite-dimensional algebra is nilpotent. Over an algebraically closed field, its semisimple quotient is a product of full matrix algebras, one for each simple module up to isomorphism. A finite-dimensional commutative algebra over an algebraically closed field is a product of local algebras with that field as residue, indexed by its primitive idempotents. Finite group algebras in characteristic zero are semisimple. Over a splitting field the simple endomorphism algebras are the scalar field, and an ordinary irreducible character satisfies

$$\sum_{x \in X} \chi(x)\chi(x^{-1}) = |X|.$$

Idempotents lift uniquely from the residue algebra of a finite commutative algebra over a complete discrete valuation ring. Nakayama’s lemma applies to finite modules over any local ring.

Foundation 3 (Commutative algebra). A finite-type algebra over a Noetherian ring is Noetherian. Integral extensions satisfy lying over. Closed points of a variety over a field are dense and have finite residue fields over the ground field. For an integral variety, the local dimension at a point plus the transcendence degree of its residue field is the transcendence degree of the function field. A regular local ring is a normal domain; a part of a minimal system of generators of its maximal ideal is a regular sequence and its quotient is regular of the correspondingly smaller dimension. A one-dimensional Noetherian normal local domain is a discrete valuation ring, as is a Noetherian local domain with nonzero principal maximal ideal. Finite torsion-free modules over discrete valuation rings are free. Completion of a discrete valuation ring retains its residue field and uniformizer.

Foundation 4 (Fields and absolute field differentials). If a finite group D acts faithfully by automorphisms on a field M , then M/M^D is Galois of degree $|D|$. Finite separable field extensions are simple. A finite field extension has a maximal separable subextension, and the remaining extension is purely inseparable; the number of embeddings in an algebraic closure is the separable degree. Multiplicative groups of finite fields are cyclic. Algebraically closed fields are classified by their characteristic and transcendence degree. If S is a finitely generated field of transcendence degree ρ over a perfect field κ , then

$$\dim_S \Omega_{S/\kappa} = \rho; \quad [S : S^p] = p^\rho \quad \text{in characteristic } p.$$

Foundation 5 (Valuations over a complete discrete base). A valuation extends to an algebraic field extension. A complete discrete valuation on a field extends uniquely to any finite field extension; its valuation ring is the integral closure of the original ring, is finite over it, and is again a complete discrete valuation ring. The fields to which we apply the latter assertion initially have characteristic zero. No corresponding finiteness assertion for an arbitrary nondiscrete valuation ring is assumed. See [37, Tag 00IA] for prolongation and [37, Tags 032W, 09EM and 0325] for the complete discrete-base finiteness assertions.

Foundation 6 (Descent and finite models). A projective scheme with a semilinear descent datum under a finite Galois field extension descends if equipped with a compatible linearized ample line bundle. The same statement holds for a finite *étale* faithfully flat Galois extension of rings and a relatively ample line bundle. Equivariant morphisms and invariant closed subschemes descend along with the scheme. Flatness, smoothness, finiteness, *étaleness*, and relative ampleness in these settings are checked after the faithfully flat extension. Geometric integrality of a finite-type scheme over a field can be checked after an extension of that field. A finite diagram of projective schemes and morphisms of finite presentation over a directed union of fields is defined over a finite stage, including its maps to a base scheme already defined there and the identities among those maps. One may realize projective descent by descent of semilinear modules and graded algebras followed by Proj. For the projective and module descent assertions, see [37, Tags 0CCJ, 0CDR and 0D1V]. The five locality assertions are respectively [37, Tags 02L2, 02VL, 02LA, 02VN and 0D3C]; geometric integrality is covered by [37, Tags 038K, 0384 and 054P]. The finite-stage assertion follows from [37, Tags 01ZM and 01ZP], including chosen projective embeddings among the finite data.

Foundation 7 (Normalization). The normalization of an integral variety over a field in a finite extension of its function field is finite. The normalization of a finite-type $\mathbb{Z}$ -domain in its fraction field is finite. In the integral closure of a normal domain in a finite Galois extension of its fraction field, the Galois group acts transitively on the primes above any fixed prime of the base. The normalization finiteness assertions follow from the Nagata property in [37, Tag 0335].

Foundation 8 (Regularity, smoothness, and properness). Finite separable scalar extension preserves regularity and reducedness. Smooth varieties over a field are regular. A regular local ring essentially of finite type over a perfect field κ is smooth at the corresponding point of a finite-type κ -model; its module of absolute differentials is free of rank the transcendence degree of the function field over κ . We use the valuative criterion for properness, including the unique extension of maps from fraction fields of valuation rings, and that proper maps are closed. For regularity and smoothness over a perfect field, see [37, Tags 00TV and 0B8X].

Foundation 9 (Proper cohomology and ample section rings). Over a Noetherian base, the higher direct images of coherent sheaves under a proper morphism are coherent. Over an affine base their cohomology commutes with flat affine base change. For a projective scheme over an affine Noetherian base and a relatively ample line bundle, sufficiently high powers give projective embeddings, and sufficiently high twists annihilate higher coherent cohomology. The ample section ring is finitely generated and its Proj recovers the scheme. Finite pullback preserves ampleness; tensor products of relatively ample line bundles are relatively ample, and a line bundle is relatively ample if a positive power is. Formation of Proj commutes with base change. On a projective curve over a field, coherent cohomology vanishes in degrees greater than one, and for sheaves of finite support in degrees greater than zero. Global functions on a geometrically integral projective variety over a field equal the ground field. For proper coherence, flat cohomology base change, and ample twists, respectively, see [37, Tags 02O5, 02KH and 0B5T]. For Proj recovery and the ampleness rules, see [37, Tags 0EKI, 0892, 0890 and 02NN].

Foundation 10 (Differentials, adjunction, and duality). We use the transitivity sequence

$$\Omega_{V/U} \otimes_V W \longrightarrow \Omega_{W/U} \longrightarrow \Omega_{W/V} \longrightarrow 0$$

and the conormal sequence for a closed immersion. Let S be a field. On $\mathbb{P}_S^b$, $\det \Omega_{\mathbb{P}_S^b/S} = \mathcal{O}(-b-1)$. For a coherent sheaf $\mathcal{G}$ there, Serre duality identifies

$$H^i(\mathcal{G})^\vee \simeq \text{Ext}^{b-i}(\mathcal{G}, \mathcal{O}(-b-1)).$$

We use the local-to-global Ext spectral sequence and the Koszul resolution of a quotient by a regular sequence. These inputs will yield the required absolute determinant-degree formula even when the curve is not smooth over S . For the projective-space dualizing complex and the adjunction formalism, see [37, Tag 0A9W and Section 48.15]. The local-to-global Ext sequence is [37, Tag 0BQP].

Foundation 11 (Elliptic curves). A smooth Weierstrass cubic with its point at infinity is an elliptic curve. A generalized Weierstrass equation over a discrete valuation ring with unit discriminant gives a smooth elliptic scheme, polarized by three times its origin. Its relative cotangent line is trivial with invariant differential η , and multiplication by an integer a pulls η back to $a\eta$. For a positive integer a , multiplication by a on an elliptic curve or elliptic scheme is finite locally free of degree a^2 . For the relative group law, see [11, II, Proposition 2.7]; the unit-discriminant Weierstrass family and the invariant cotangent description are given in [37, Tags 072S and 047I]. See also [34, Chapter III] for the Weierstrass formulation. Multiplication is finite flat and surjective by [30, Proposition 20.7]; over the Noetherian bases here it is finite locally free by [37, Tag 02KB].

Foundation 12 (Complex curves and covers). Finite étale algebraic covers of a smooth complex algebraic curve correspond to finite unramified topological covers of its analytification, including morphisms. For a punctured projective curve, normalization of the original projective curve in the resulting algebraic function field gives the finite compact cover. Normal complex curves are smooth, and the local punctured covering completes by a power map, so monodromy cycle lengths give ramification indices. Topological covers are classified by fundamental-group actions on finite fibers. The fundamental group of a genus-one surface punctured at n points has generators $x, y, u_1, \dots, u_n$ with the single relation

$$[x, y]u_1 \cdots u_n = 1.$$

For a finite map of smooth proper curves over an algebraically closed field of characteristic zero, Riemann–Hurwitz states

$$2g(C) - 2 = \deg(f)(2g(C') - 2) + \sum_{P \in C} (e_P - 1).$$

For the algebraic/topological cover correspondence, see [29, Theorem 3.4]; for Riemann–Hurwitz, see [37, Section 53.12, Tag 0C1B].

Foundation 13 (Noetherian connectedness). A proper morphism over a locally Noetherian base whose structure-sheaf direct image is canonically the structure sheaf of the base has geometrically connected fibers. This is the connected-fiber assertion of Stein factorization. We apply it only after constructing a Noetherian normal model, not directly over the nondiscrete valuation ring. See [37, Tag 03H0].

2.1 Coefficient fields and scalar extension

We justify the coefficient choice in [Section 1](#) without requiring a mixed-characteristic lift of every algebraically closed field. Put $\kappa = \overline{\mathbb{F}}_p$ and choose an embedding $\kappa \hookrightarrow k$. For any finite group X , the ideal $\text{rad}(\kappa X) \otimes_{\kappa} k$ is nilpotent, and the quotient of kX by this ideal is the scalar extension of the split matrix product $\kappa X / \text{rad}(\kappa X)$. That quotient is again a split semisimple algebra. The ideal is therefore exactly $\text{rad}(kX)$, and the simple modules correspond under scalar extension.

The center also commutes with this extension: it is the kernel of the finitely many linear maps $a \mapsto ax - xa$, $x \in X$, and field extension preserves kernels. Each local factor of $Z(\kappa X)$ has nilpotent radical and residue κ ; after extension it still has nilpotent radical, now with residue k , so remains local. Thus primitive blocks correspond and their simple-module counts agree. Coefficient restriction, restriction to centralizers in the Brauer map, and the quotient maps for normal subgroups all commute with scalar extension. The residue characters of the local central factors extend as well. Consequently the central-character block assignment in [\(1.1\)](#) is preserved, for every subgroup and subquotient involved in a weight. It suffices to prove the numerical identity over κ .

To construct the coefficient ring, start with $\mathbb{Z}_p$, lift a tower of finite separable extensions of its residue field $\mathbb{F}_p$ that exhausts κ , and complete the union. The construction at the start of the proof of [Proposition 6.1](#) shows that this gives a complete discrete valuation ring of characteristic zero with residue κ . In an algebraic closure of its fraction field, choose matrix realizations of every ordinary irreducible representation of every subgroup of G . There are only finitely many such representations and matrix entries. Adjoining their entries gives a finite field extension that splits all those subgroups. Its valuation ring is again a complete DVR by [Foundation 5](#); its finite residue extension is still κ , which is algebraically closed. This gives the system $(K, \mathcal{O}, \kappa)$ used in [Section 3](#). Quotient representations inflate, so the same coefficients split the required subquotients. These coefficient fields are separate from the auxiliary geometric fields constructed later.

The remaining sections prove the blockwise chain identity, the marked high-index twist, the genus estimate with exact rounding, the required valued-field assertions, and the specialization model. The foundations above supply their classical inputs, not the desired weight equality.

3 From block counts to generating tuples

Fix a prime p . Write $O_p(X)$ for the largest normal p -subgroup of a finite group X , and put $[x, y] = xyx^{-1}y^{-1}$. A p -element includes the identity; a p' -element has order prime to p ; and a p -singular element has order divisible by p . The geometric sections will establish the following precise counting statement. Its uniformity is part of the assertion.

Theorem 3.1 (Uniform divisibility for generating tuples). *Let J be a finite group with $O_p(J) = 1$. For $s \geq 0$, $n = p^s$, and a multiset $\mathcal{M}$ of n conjugacy classes of p -singular elements of J , let $N_{J,s}(\mathcal{M})$ be the number, modulo simultaneous J -conjugation, of ordered tuples*

$$(x, y, u_1, \dots, u_n) \in J^{n+2}$$

satisfying

$$[x, y]u_1 \cdots u_n = 1, \quad \langle x, y, u_1, \dots, u_n \rangle = J, \quad \{[u_1], \dots, [u_n]\} = \mathcal{M}. \quad (3.1)$$

For every integer $h \geq 1$, there is an integer $s_0 = s_0(p, J, h)$ such that

$$p^h \mid N_{J,s}(\mathcal{M}) \quad (s \geq s_0)$$

for every such multiset $\mathcal{M}$.

The entries u_i remain ordered: prescribing their multiset of classes does not quotient by permutations. Empty tuple sets have count zero, so the statement includes $J = 1$ and groups of order prime to p . The proof of [Theorem 3.1](#) is completed after [Proposition 7.1](#). This section proves that it implies the block equality, using [Foundations 1](#) and [2](#) and proving the required counting reductions explicitly.

3.1 Coefficient projection and the weight transform

For each ambient finite group X , choose a splitting coefficient system $(K, \mathcal{O}, k)$ as in [Sections 1](#) and [2.1](#), with X in place of G . Hold this system fixed throughout the calculations with X and its subquotients. This system also splits all subquotients that occur below: a subgroup of a quotient lifts to a subgroup by taking its preimage, and a representation of a quotient inflates to that preimage. These observations apply after any number of subgroup and quotient steps. Ordinary irreducibles may therefore be realized over K . Their identification with complex irreducibles, when needed, follows by first splitting the group algebra over $\mathbb{Q}$ and then extending scalars to algebraically closed characteristic-zero fields.

For a central idempotent $E \in kX$, define

$$\begin{aligned} l(X, E) &= \#\{\text{simple } kX\text{-modules selected by } E\}, \\ k_{\text{ord}}(X, E) &= \#\{\text{ordinary irreducibles selected by the lift of } E\}, \\ z(X, E) &= \#\{\chi \text{ selected by that lift} : \chi(1)_p = |X|_p\}. \end{aligned}$$

Here and throughout, modules and characters are counted up to isomorphism. The class sums form a basis of $Z(\mathcal{O}X)$, and their reductions form a basis of $Z(kX)$. Idempotent lifting over the complete DVR thus gives a unique central idempotent lifting E . Every integral central element acts on an ordinary irreducible by a scalar in $\mathcal{O}$: the scalar lies in K by splitting and satisfies a monic integral equation by finiteness, and $\mathcal{O}$ is integrally closed. Reduction of this scalar defines a character of $Z(kX)$. The Artinian local decomposition of that center shows that it is precisely λ_b , where b is the block containing the ordinary irreducible.

For $H \leq X$, let s_H denote coefficient restriction from $Z(kX)$ to $Z(kH)$, or its integral analogue when appropriate. For a p -subgroup Q , let Br_Q denote restriction of coefficients from the Q -invariants $(kX)^Q$ to $kC_X(Q)$.

Lemma 3.2 (Normalizer projection). *The map $\text{Br}_Q : (kX)^Q \rightarrow kC_X(Q)$ is a unital algebra homomorphism. Suppose that*

$$Q \leq H \leq N_X(Q), \quad C_X(Q) \leq H.$$

Writing $\pi_Q : kH \rightarrow k[H/Q]$ for the quotient map, one has, for $v \in Z(kX)$,

$$\pi_Q(s_H(v)) = \pi_Q(\text{Br}_Q(v)), \quad \lambda_b(s_H(v)) = \lambda_b(\text{Br}_Q(v)) \tag{3.2}$$

for every block b of H . In particular $\lambda_b \circ s_H$ is the central character of a unique block of X .

Proof. To compute the coefficient of $g \in C_X(Q)$ in a product of two Q -invariant elements, let Q conjugate the pairs (a, b) with $ab = g$. Coefficient products are constant on the orbits. Every nontrivial orbit has size divisible by p , and the fixed pairs are exactly those with $a, b \in C_X(Q)$. This proves multiplicativity; the identity is preserved.

Each coset hQ is stable under conjugation by Q , because $h \in N_X(Q)$. Applying the same orbit cancellation to the sum of coefficients over that coset proves the first identity in (3.2). Also $\text{Br}_Q(v) \in Z(kH)$, since its support lies in $C_X(Q) \leq H$ and it is invariant under H .

Every nonzero module for a finite p -group over k has nonzero invariants. For completeness, choose a central element z of order p ; the nonzero kernel of $z - 1$, whose p th power is zero, is a module for the smaller group $Q/\langle z \rangle$, and induction applies. If the module is simple for H , its Q -invariants are H -stable, and hence the whole module. Thus every simple kH -module factors through H/Q , proving the second identity. The composition with Br_Q is a unital algebra character. Characters of the finite commutative algebra $Z(kX)$ are its local residue characters, one for each block, giving uniqueness. $\square$

For $H = N_X(Q)$ put

$$\overline{E}_Q = \pi_Q(s_H(E)) = \pi_Q(\text{Br}_Q(E)) \in Z(k[H/Q]).$$

This is an idempotent. Its selected defect-zero quotient characters are exactly the weights assigned to the blocks appearing in E . Indeed, the lift of $\text{Br}_Q(E)$ maps to the lift of $\overline{E}_Q$ by uniqueness of central idempotent lifting. Its action on the inflation of a quotient character is therefore one exactly when the latter is selected by $\overline{E}_Q$. If b is the block of this inflation, the same test is

$$\lambda_b(\text{Br}_Q(E)) = \lambda_b(s_H(E)) = 1.$$

By [Lemma 3.2](#), this says that the induced block is one of the blocks in E , with the coefficient-projection definition in [\(1.1\)](#). Once Q is fixed up to X -conjugacy, there is no further identification of its quotient characters: $N_X(Q)$ acts on $N_X(Q)/Q$ by inner automorphisms.

For a conjugation-invariant function g of such pairs (X, E) , define

$$(Tg)(X, E) = \sum_{\substack{1 < Q \leq X \text{ a } p\text{-subgroup} \\ \text{up to } X\text{-conjugacy}}} g(N_X(Q)/Q, \overline{E}_Q). \quad (3.3)$$

Consequently the desired universal equality for block sums is $l = (1 + T)z$. The identity term includes precisely $Q = 1$. Every transition in T strictly decreases the p -part of the group order, so T is nilpotent on each evaluated pair. Thus the formal inverse $(1 + T)^{-1} = \sum_{r \geq 0} (-T)^r$ is evaluation-wise finite.

The next step makes the connection with subgroup-chain formulations of the weight conjecture [\[25\]](#). We derive the particular normal-chain identity from this finite inverse, retaining its idempotent and orbit data throughout.

Definition 3.3 (Chains). A chain in X is a strictly increasing sequence of p -subgroups

$$\mathcal{C} = (1 = Q_0 < Q_1 < \cdots < Q_r), \quad |\mathcal{C}| = r.$$

The length-zero chain (1) is included. Set

$$H_{\mathcal{C}} = \bigcap_{i=0}^r N_X(Q_i), \quad E_{\mathcal{C}} = \text{Br}_{Q_r}(E).$$

Call $\mathcal{C}$ *normal* if every Q_i is normal in Q_r .

For every chain, normal or otherwise, $C_X(Q_r) \leq H_{\mathcal{C}} \leq N_X(Q_r)$, so $E_{\mathcal{C}}$ is an idempotent of $Z(kH_{\mathcal{C}})$. For a normal chain one also has $Q_r \leq H_{\mathcal{C}}$.

Lemma 3.4 (Finite chain inversion). *The identity $l = (1 + T)z$ for all pairs (X, E) is equivalent to*

$$z(X, E) = \sum_{\substack{\mathcal{C}/X \\ \mathcal{C} \text{ normal}}} (-1)^{|\mathcal{C}|} l(H_{\mathcal{C}}, E_{\mathcal{C}}), \quad (3.4)$$

where $\mathcal{C}/X$ denotes one representative of each X -conjugacy orbit.

Proof. We identify the terms of $T^r l$. After r steps they correspond to normal chains of length r , with group H_C/Q_r and idempotent

$$\pi_{Q_r}(s_{H_C}(E)) = \pi_{Q_r}(E_C).$$

For the induction, the preimage Q_{r+1} of a nontrivial p -subgroup of H_C/Q_r is a p -group strictly containing Q_r . It lies in H_C and therefore normalizes every preceding Q_i . Conversely every normal one-step extension arises in this way. Its normalizer in the quotient has preimage $H_C \cap N_X(Q_{r+1}) = H_{C'}$. After fixing C , conjugation is by its stabilizer H_C , which is exactly quotient conjugation at the next step. This proves the orbit assertion without an extra multiplicity.

To check the idempotent at the next step, write $E = \sum_{x \in X} e_x x$ and put $H' = H_{C'}$. The new normalizer in H_C/Q_r is H'/Q_r . Restriction to this normalizer retains exactly the coefficient sums over the Q_r -cosets contained in H' . Choose a set A of representatives for the cosets of Q_r in Q_{r+1} . For $h \in H'$, the subsequent quotient has coefficient

$$\sum_{a \in A} \sum_{q \in Q_r} e_{haq} = \sum_{t \in Q_{r+1}} e_{ht}$$

at hQ_{r+1} : the sets aQ_r , $a \in A$, partition Q_{r+1} . These are precisely the coefficients of $\pi_{Q_{r+1}}(s_{H'}(E))$. By Lemma 3.2, this idempotent is $\pi_{Q_{r+1}}(E_{C'})$, as required. Every simple kH_C -module factors through H_C/Q_r and has the indicated idempotent action. Hence the contribution to $T^r l$ is the corresponding term in (3.4). The finite inverse of $1 + T$ now gives the equivalence. $\square$

3.2 Cocenters and subsections

By Lemma 3.4, the remaining algebraic objective is (3.4). We next express ordinary character counts as sums of simple-module counts in centralizers. Chain cancellation will then replace the alternating sum of l by an alternating sum of k_{ord} , which can be computed by central traces.

For an algebra A , write $V(A) = A/[A, A]$, where $[A, A]$ is the linear span of commutators, and write $\bar{a}$ for the image of a . A central element acts on $V(A)$ by multiplication.

Powers modulo commutators underlie Külshammer's generalized Reynolds ideals; see [18, Section 2]. We give the finite-dimensional argument needed to extract the two block counts.

Lemma 3.5 (Cocenter ranks). *For a finite-dimensional k -algebra A , the map*

$$F_A : V(A) \longrightarrow V(A), \quad \bar{a} \longmapsto \overline{a^p}$$

is well-defined and Frobenius semilinear. Its stable image has dimension equal to the number of simple A -modules, and its summand selected by a central idempotent has dimension equal to the corresponding simple-module count. For $A = kX$, this stable image is the span of the p' -classes. The ranks of multiplication by E on that span and on all of $V(kX)$ are, respectively, $l(X, E)$ and $k_{\text{ord}}(X, E)$.

Proof. In the expansion of $(a + b)^p$, cyclic rotations of any mixed word give the same cocenter class. Their orbits have size p , leaving only a^p and b^p . Thus the power operation into the cocenter is additive and is Frobenius semilinear. Moreover $\overline{(ab)^p} = \overline{(ba)^p}$, by one cyclic rotation; additivity shows that this operation kills the linear commutator space. It therefore descends to F_A .

Let R be the Jacobson radical of A . The kernel of $V(A) \longrightarrow V(A/R)$ is represented by elements of R : lift any commutator expression downstairs and subtract it upstairs. Since R is nilpotent, a sufficiently high iterate of F_A kills this kernel. On the split semisimple algebra A/R , the cocenter

has one matrix-trace coordinate per simple factor, and F_A raises each coordinate to the p th power. It is bijective because k is perfect. For example, the trace identity here follows by triangularizing a matrix over k and taking the p th powers of its diagonal entries.

The descending images of F_A stabilize, since they are k -subspaces of a finite-dimensional space. Denote the stable image by S and the quotient map by $q : V(A) \rightarrow V(A/R)$. Choose m large enough that $S = \text{im } F_A^m$ and F_A^m kills $\ker q$. On S , the map F_A is surjective and hence bijective, because it is semilinear for the automorphism $a \mapsto a^p$ of the perfect field k . If $x \in S \cap \ker q$, then $F_A^m(x) = 0$; injectivity on S gives $x = 0$. To prove surjectivity of $q|_S$, take $y \in V(A/R)$. The Frobenius map $F_{A/R}$ is bijective, so choose y' with $F_{A/R}^m(y') = y$ and lift y' to $x \in V(A)$. Then $F_A^m(x) \in S$ and $q(F_A^m(x)) = y$. Thus $q|_S$ is an isomorphism. Finally $(Ea)^p = Ea^p$ for a central idempotent E , so all these statements respect its summand.

For a group algebra the cocenter is free on conjugacy classes, represented by individual group elements: the commutator relations identify exactly conjugate elements. Sufficiently high p -powers land on p' -elements, and taking p th powers permutes the p' -classes. This identifies the stable image. The integral cocenter $V(\mathcal{O}X)$ is free on the same class basis. The lift of E projects it onto a free direct summand, whose rank is unchanged over k and K . Split semisimplicity over K identifies this rank with $k_{\text{ord}}(X, E)$. $\square$

Every group element has a unique commuting factorization into its p -part and p' -part, both powers of that element. A *section* is the set of elements whose p -parts are conjugate to a fixed p -element.

The next identity is Brauer's classical subsection count [5]. The cocenter argument also proves the support property needed for the defect-zero detector, without dividing by class sizes in characteristic p .

Lemma 3.6 (Subsection formula and support). *For every central idempotent $E \in kX$,*

$$k_{\text{ord}}(X, E) = \sum_{\substack{[u] \text{ an } X\text{-class} \\ u \text{ a } p\text{-element}}} l(C_X(u), \text{Br}_{\langle u \rangle}(E)). \quad (3.5)$$

Multiplication by E preserves the section decomposition of the cocenter and the section decomposition of the center into spans of class sums.

Proof. Fix a p -element u and set $C = C_X(u)$. The map $\bar{t} \mapsto \overline{ut}$ identifies the p' -class span of $V(kC)$ with the section span belonging to u in $V(kX)$. It is a bijection of class bases: conjugacy of ut and ut' forces a conjugating element to centralize their common p -part u , and the converse is immediate.

For $t \in C$ a p' -element, conjugation by $\langle u \rangle$ in the expansion of Eut gives

$$\overline{Eut} = \overline{\text{Br}_{\langle u \rangle}(E)ut}.$$

Indeed, the coefficients are constant on these orbits and the cocenter classes of their products with ut agree; only fixed orbits survive. The restricted idempotent is central in kC and preserves its p' -class span by Lemma 3.5. Multiplication by u is central in kC , so the displayed identification intertwines the two idempotent actions. Their ranks are thus the terms in (3.5). Summing the ranks over the sections proves that identity.

For the center, use the pairing

$$Z(kX) \times V(kX) \longrightarrow k, \quad (a, \bar{b}) \longmapsto \tau(ab),$$

where τ takes the coefficient of 1. It is nondegenerate: a class sum pairs with an individual representative of its inverse class with coefficient one. Multiplication by E is self-adjoint for this pairing. Sections on the center pair with the inverse sections of the cocenter, so preservation there proves preservation here. In particular no class-size division is involved. $\square$

3.3 Two cancellations on chain orbits

Lemma 3.7 (Removal of nonnormal chains). *In an alternating sum over chain orbits, every contribution that depends only on H_C and Q_r cancels on the nonnormal chains. Thus such a sum may be taken over all chains or only normal chains.*

Proof. For a nonnormal chain, let i be the largest index for which $A = Q_i$ is not normal in $P = Q_r$, and let R be the normal closure of A in P . Then $A < R \leq Q_{i+1}$: every later member is normal in P . Also $R < P$. To see the latter, put the proper subgroup A in a maximal subgroup of P , which is normal because P is a finite p -group; it contains R .

Insert R immediately after A if it is absent, and remove it if it is present. The last group P and the last nonnormal member A are unchanged, so this operation is an involution. Every element normalizing A and P normalizes their normal closure R ; hence insertion or removal leaves H_C unchanged. The operation commutes with X -conjugation and changes the length by one. It induces a fixed-point-free sign reversal on the orbits, which proves cancellation. $\square$

Lemma 3.8 (Alternating subsection cancellation). *One has*

$$\sum_{\substack{\mathcal{C}/X \\ \mathcal{C} \text{ normal}}} (-1)^{|\mathcal{C}|} l(H_C, E_C) = \sum_{\substack{\mathcal{C}/X \\ \mathcal{C} \text{ normal}}} (-1)^{|\mathcal{C}|} k_{\text{ord}}(H_C, E_C). \quad (3.6)$$

Proof. By Lemma 3.7, both sums may be taken over all chains. Expand the ordinary count by Lemma 3.6. The $u = 1$ terms give the simple-module sum. The other terms are indexed by X -orbits of pairs $(\mathcal{C}, u)$, with $u \neq 1$ a p -element in H_C . This orbit description follows because the stabilizer of a fixed chain is H_C . The contribution of such a pair is

$$l(H_C \cap C_X(u), \text{Br}_{Q_r\langle u \rangle}(E)). \quad (3.7)$$

Here u normalizes Q_r , so $Q_r\langle u \rangle$ is a p -group; successive coefficient restrictions have support $C_X(Q_r) \cap C_X(u) = C_X(Q_r\langle u \rangle)$, proving the idempotent in this expression.

Let i be the largest index, allowing 0 and r , for which $u \notin Q_i$. Such an index exists since $Q_0 = 1$. Toggle $R = Q_i\langle u \rangle$ immediately after Q_i : insert it if absent, and remove it if present. It is a p -group, since u normalizes Q_i , and it is contained in every later member, all of which contain u . The last member not containing u therefore remains Q_i ; this proves involutivity, including an insertion after the terminal member and its inverse deletion.

Every element centralizing u and normalizing Q_i normalizes R . Thus $H_C \cap C_X(u)$ is unchanged. The product of the terminal member with $\langle u \rangle$ is also unchanged, even when the terminal member is inserted or deleted. Hence (3.7) is preserved. The operation is X -equivariant and reverses the sign, so all $u \neq 1$ terms cancel. $\square$

3.4 Detecting defect zero by central elements

By Lemma 3.8, it remains to prove that the alternating ordinary-character sum in (3.6) equals $z(X, E)$. Their difference retains the signed contributions of all characters selected by E_C on positive-length normal chains, but omits the selected defect-zero characters at the chain (1). We will realize this difference as a trace limit by constructing a central element with the corresponding zero-or-one central-character values.

Put

$$c_X = \sum_{u \in X \text{ a } p\text{-element}} u, \quad d_X = \sum_{x, y \in X} [x, y], \quad I_X(a) = \sum_{x \in X} xax^{-1}.$$

These elements and the image of I_X are central. Call a block *semisimple* if its algebra is semisimple; since a block has no nontrivial central idempotent, such a block is one full matrix algebra over k .

The scalar of d_X will identify these semisimple blocks with the ordinary defect-zero characters. The complement $1 - c_X$ will then remove exactly those blocks from the central trace, using the support information supplied by I_X .

The operator I_X is the group-algebra Higman trace [19, Section 2]; see also [27, Sections 2.4–2.5]. The use of c_X to isolate defect zero goes back to Tsushima [38, Theorem 1]. The following proof records the exact radical-annihilation and central-character consequences used in our trace limit.

Lemma 3.9 (Averaging and defect zero). *The image of I_X , and also d_X , annihilates the Jacobson radical of kX . On a semisimple block, d_X has nonzero central scalar; on every other block its central character is zero. Each semisimple block contains exactly one ordinary irreducible, which has defect zero, and every ordinary defect-zero irreducible occurs in this way. Moreover:*

- (i) $\text{im } I_X$ is the span of class sums of elements whose centralizers have order prime to p ;
- (ii) each semisimple block idempotent belongs to $\text{im } I_X$ and has zero Brauer image at every nontrivial p -subgroup;
- (iii) $\lambda_b(c_X) = 0$ on every nonsemisimple block b .

Proof. Radical annihilation. The form $(a, b) \mapsto \tau(ab)$ on kX is symmetric and nondegenerate. For dual bases v_i, v_i^* , the tensor $\Omega = \sum_i v_i \otimes v_i^*$ is independent of the chosen bases. Apply the linear map $u \otimes v \mapsto uav$ to this tensor. In the group basis, whose dual basis consists of the inverses, the result is $I_X(a)$. Similarly, apply $u \otimes v \otimes w \otimes z \mapsto u w v z$ to $\Omega \otimes \Omega$. In the group basis the result is $\sum_{x, y \in X} x y x^{-1} y^{-1} = d_X$. Thus in any dual bases

$$I_X(a) = \sum_i v_i a v_i^*, \quad d_X = \sum_{i, j} v_i v_j v_i^* v_j^* = \sum_j I_X(v_j) v_j^*. \quad (3.8)$$

For b in the radical R , the trace identity

$$\tau(I_X(a)b) = \text{Tr}(L_b R_a)$$

follows by writing the trace in these dual bases; L and R denote left and right multiplication. Since L_b is nilpotent and commutes with R_a , this trace is zero. Replace b by bc for arbitrary $c \in kX$, and use nondegeneracy to obtain $I_X(a)b = 0$. The final expression in (3.8) then shows that $d_X R = 0$ as well, because $v_j^* R \subseteq R$.

Semisimple blocks and ordinary defect zero. Distinct block ideals are orthogonal for the form. On a semisimple block $M_d(k)$ its restriction is $t \text{Tr}$ for some $t \in k^\times$. Indeed a symmetric functional kills commutators, whose quotient in a matrix algebra is the trace line, and nondegeneracy forces $t \neq 0$. Matrix units e_{ij} have duals $t^{-1} e_{ji}$. Since $\sum_{i, j} e_{ij} a e_{ji} = \text{Tr}(a) 1$, the first contraction in (3.8) gives $I_X(a) = t^{-1} \text{Tr}(a) 1$. In particular $I_X(e_{ij}) = t^{-1} \delta_{ij} 1$, so its last expression gives

$$d_X = \sum_{i, j} I_X(e_{ij}) t^{-1} e_{ji} = t^{-2} \sum_i e_{ii} = t^{-2} 1.$$

We have therefore proved

$$I_X(a) = t^{-1} \text{Tr}(a) 1, \quad d_X = t^{-2} 1 \quad \text{on this block.} \quad (3.9)$$

In particular its idempotent lies in the image of I_X ; one can take $a = t e_{11}$. These formulas do not require d to be nonzero in k . On a nonsemisimple block, a central element with nonzero residue

scalar is a unit in its local center. Such a unit cannot annihilate the nonzero radical. This proves the assertion about the central character of d_X .

Now compute the same commutator sum in characteristic zero. If χ has degree d , Schur's lemma gives

$$\sum_{x \in X} \rho(xy x^{-1}) = \frac{|X| \chi(y)}{d} 1.$$

Multiply by $\rho(y^{-1})$, sum over y , and take traces. Ordinary character orthogonality gives the scalar

$$\lambda_\chi(d_X) = \frac{|X|^2}{\chi(1)^2}. \quad (3.10)$$

It is integral, by the integrality of ordinary central scalars. Its reduction is nonzero exactly when $v_p(\chi(1)) = v_p(|X|)$, namely in defect zero. The modular calculation therefore puts exactly these characters in semisimple blocks. Such a block has cocenter dimension one, so [Lemma 3.5](#) shows that its lifted block selects exactly one ordinary irreducible. This also proves existence of that character.

Support and the p -element sum. For a group element g one has

$$I_X(g) = |C_X(g)| \sum_{a \in [g]} a.$$

This proves assertion (i). The support of a semisimple block idempotent, already in this image, therefore contains no element centralized by a nontrivial p -subgroup, proving (ii).

Finally fix a nontrivial p -subgroup P . Choose $z \in Z(P)$ of order p . It is central also in $C_X(P)$, and multiplication by z preserves the set of p -elements of that centralizer. With $q = \sum_{j=0}^{p-1} z^j$, the sum of those p -elements factors as q times a sum of orbit representatives. As q is central and $q^2 = pq = 0$, this proves

$$\text{Br}_P(c_X)^2 = 0, \quad \text{Br}_P(c_X^2) = 0.$$

If p divides $|C_X(g)|$, a nontrivial p -subgroup $P \leq C_X(g)$ makes the coefficient of g in c_X^2 vanish by this identity. Assertion (i) gives $c_X^2 \in \text{im } I_X$. On a nonsemisimple block its central character must therefore vanish, by the same radical-annihilator argument. Since $\lambda_b(c_X)^2 = \lambda_b(c_X^2) = 0$, assertion (iii) follows. $\square$

Lemma 3.10 (An element with zero-or-one residues). *Let $v = E(1 - c_X)$. It is central and supported on p -singular elements. For a normal chain C of positive length, its restriction to H_C satisfies*

$$\lambda_b(s_{H_C}(v)) = \lambda_b(E_C) \in \{0, 1\}$$

for every block b of H_C . For the length-zero chain, the scalar is one on the nonsemisimple blocks selected by E and zero on every other block.

Proof. The element $1 - c_X$ is the negative sum of the nonidentity p -elements, hence lies entirely in p -singular sections. Section preservation from [Lemma 3.6](#) proves the same support assertion for v . Let B_0 be a semisimple block idempotent. Since v is central and $B_0 kX$ is a full matrix algebra, $B_0 v$ lies in its one-dimensional center kB_0 . On the other hand, section preservation makes the support of $B_0 v$ p -singular. By [Lemma 3.9](#), the support of B_0 contains only elements whose centralizers have order prime to p , and therefore no p -singular elements. The two support conditions force the scalar multiple $B_0 v$ to be zero. On a nonsemisimple block, [Lemma 3.9](#) gives $\lambda_b(v) = \lambda_b(E)$, proving the length-zero assertion.

For positive length, put $Q = Q_r \neq 1$ and $H = H_C$. By [Lemma 3.2](#), $\lambda_b \circ s_H$ is a central character of X and factors through Br_Q . It cannot be the character of a semisimple block B_0 : that would give value one on B_0 , whereas $\text{Br}_Q(B_0) = 0$ by [Lemma 3.9](#). On its nonsemisimple block, $\lambda(v) = \lambda(E)$. Applying [Lemma 3.2](#) to E gives the asserted value $\lambda_b(E_C)$. $\square$

3.5 Central traces and the final cancellation

Choose a class-sum lift

$$\hat{v} = \sum_{a \in X} \alpha_a a \in Z(\mathcal{O}X)$$

of the detecting element $v = E(1 - c_X)$ from Lemma 3.10, with $\alpha_a = 0$ whenever a is not p -singular. For a positive integer n , define

$$\Phi_n = \sum_{\mathcal{C}/X} (-1)^{|\mathcal{C}|} \text{Tr}(s_{H_{\mathcal{C}}}(\hat{v})^n \mid V(KH_{\mathcal{C}})), \quad (3.11)$$

where the operator is central multiplication and all chains are included.

Lemma 3.11 (The first trace limit). *In the valuation topology of K ,*

$$\lim_{s \rightarrow \infty} \Phi_{p^s} = \sum_{\substack{\mathcal{C}/X \\ \mathcal{C} \text{ normal}}} (-1)^{|\mathcal{C}|} k_{\text{ord}}(H_{\mathcal{C}}, E_{\mathcal{C}}) - z(X, E). \quad (3.12)$$

Proof. The nonnormal chains cancel by Lemma 3.7, since their trace contribution depends only on the stabilizer. For a remaining chain, the cocenter of $KH_{\mathcal{C}}$ has one coordinate per ordinary irreducible. The unpowered operator has on these coordinates integral central eigenvalues reducing to the scalars in Lemma 3.10.

If an eigenvalue reduces to zero, its p^s th powers tend to zero. If it is $1 + a$ with $\text{val}(a) > 0$, the binomial formula gives

$$\text{val}((1 + a)^p - 1) \geq \min\{\text{val}(p) + \text{val}(a), p \text{val}(a)\}.$$

After normalizing the discrete valuation to integer values, each iteration increases the positive valuation by at least one, unless the difference has already become zero. These powers tend to one. This argument applies also to $p = 2$ and requires no finiteness of the residue field.

For positive-length chains the limiting trace therefore counts all ordinary irreducibles selected by $E_{\mathcal{C}}$. At length zero it omits the semisimple blocks, each of which contributes exactly one defect-zero character by Lemma 3.9. There are finitely many chains and irreducibles, so limits may be summed, proving (3.12). $\square$

The next identity follows from Frobenius's commutator-counting formula and is the weighted genus-one case of the surface formula in [24, Theorem 2.1 and Equation (3.1)]. Its single denominator is important when chain orbits are replaced by actual chains, so we verify the normalization directly.

Lemma 3.12 (Trace normalization). *For $H \leq X$ and $w = s_H(\hat{v})$,*

$$\text{Tr}(w^n \mid V(KH)) = \frac{1}{|H|} \sum_{\substack{x, y, u_1, \dots, u_n \in H \\ [x, y] u_1 \cdots u_n = 1}} \prod_{i=1}^n \alpha_{u_i}. \quad (3.13)$$

Proof. The sum on the right before division is $\tau(d_H w^n)$. The regular trace of any element of KH is $|H|$ times its identity coefficient. On the other hand, (3.10) and the regular decomposition give

$$\begin{aligned} |H| \tau(d_H w^n) &= \sum_{\chi \in \text{Irr}(H)} \chi(1)^2 \frac{|H|^2}{\chi(1)^2} \lambda_{\chi}(w)^n \\ &= |H|^2 \text{Tr}(w^n \mid V(KH)). \end{aligned}$$

Dividing proves exactly the single denominator in (3.13). $\square$

For a subgroup $J \leq X$, define an integer on actual chains, without quotienting by conjugation,

$$a_X(J) = \sum_{\substack{\mathcal{C} \text{ a chain in } X \\ J \leq H_{\mathcal{C}}}} (-1)^{|\mathcal{C}|}. \quad (3.14)$$

Lemma 3.13 (Cancellation by a normal p -subgroup). *If $O_p(J) \neq 1$, then $a_X(J) = 0$.*

Proof. Put $P = O_p(J)$. In any chain normalized by J , each Q_i is normalized by $P \leq J$, so $Q_i P$ is a p -subgroup. Both factors are normalized by J , making their product J -normalized. Let i be the last index such that $P \not\leq Q_i$; it exists since $Q_0 = 1$ and $P \neq 1$. Toggle $Q_i P$ immediately after Q_i . It strictly contains Q_i and is contained in all later members. Insertion and deletion preserve the last member not containing P , so this is an involution. It is defined also for the length-zero chain and for terminal insertion or deletion. It preserves the property of being normalized by J and reverses the sign. The sum of actual chains in (3.14) therefore vanishes. $\square$

Proposition 3.14 (Reduction to uniform tuple divisibility). *If Theorem 3.1 holds for every finite group with trivial p -core, then for every finite group X and central idempotent $E \in kX$ one has $l(X, E) = ((1 + T)z)(X, E)$. In particular the block equality in Theorem 1.1 follows, with the stated block assignment and with $Q = 1$ included.*

Proof. The orbit of a chain $\mathcal{C}$ has size $|X|/|H_{\mathcal{C}}|$. Consequently Lemma 3.12 changes the orbit sum (3.11) into the following sum over actual tuples and actual chains:

$$\Phi_n = \frac{1}{|X|} \sum_{\substack{x, y, u_1, \dots, u_n \in X \\ [x, y] u_1 \dots u_n = 1}} a_X(\langle x, y, u_1, \dots, u_n \rangle) \prod_{i=1}^n \alpha_{u_i}. \quad (3.15)$$

To verify the factor, each orbit contribution has prefactor $1/|H_{\mathcal{C}}|$; replacing its representative by all $|X|/|H_{\mathcal{C}}|$ conjugates gives the common prefactor $1/|X|$. For a fixed tuple the admissible chains are exactly those normalized by its generated subgroup. This is precisely $a_X(J)$.

By Lemma 3.13, only generated subgroups J with $O_p(J) = 1$ contribute. By the support of $\hat{v}$, only tuples with every u_i p -singular contribute. Fix such a subgroup J and group its generating tuples according to the multiset $\mathcal{M}$ of J -conjugacy classes of the u_i . The product $\prod_i \alpha_{u_i}$ is constant on this multiset: the coefficients are constant on X -classes, hence on J -classes, and their product is unchanged by reordering. Denote this integral product by $\alpha(\mathcal{M})$. A generating tuple has simultaneous conjugation stabilizer $Z(J)$, since an element centralizing all its entries centralizes the group they generate. Thus its number of actual tuples is $|J|/|Z(J)|$ times its orbit count. For $n = p^s$, the numerator in (3.15) is therefore

$$\sum_{\substack{J \leq X \\ O_p(J) = 1}} a_X(J) \frac{|J|}{|Z(J)|} \sum_{\mathcal{M}} \alpha(\mathcal{M}) N_{J,s}(\mathcal{M}). \quad (3.16)$$

Fix any $h \geq 1$. By Theorem 3.1, each summand of the inner sum belongs to $p^h \mathcal{O}$ for all sufficiently large s , with a threshold independent of $\mathcal{M}$. Although the number of multisets grows with s , the sum is finite for each s and remains in that same ideal. There are only finitely many subgroups $J \leq X$, so take the largest of their thresholds. The integer factors in (3.16) preserve divisibility. We conclude that its numerator tends to zero in the valuation topology. The fixed denominator $|X|$ does not change this conclusion, so $\Phi_{p^s} \rightarrow 0$.

Combine this with Lemma 3.11 and Lemma 3.8. The resulting equality is (3.4); its integer terms agree in the characteristic-zero field K , and hence agree as integers. The same reasoning applies to every pair and to the splitting subquotients arising in the transform. Lemma 3.4 now gives $l = (1 + T)z$. The weight interpretation of (3.3) proves the asserted block equality. $\square$

4 A genus-one curve with large index and independent labels

The fields in this section are auxiliary to the block coefficient system. Fix a prime p and put $\kappa = \overline{\mathbb{F}}_p$. For $n = p^s$ we will construct a genus-one curve E/F_0 with a finite étale divisor $\mathcal{B}$ of degree n . Every divisor on E will have degree divisible by n . We will also choose a constant elliptic curve A_0/κ and a cyclic extension F'/F_0 with $F' = \kappa(A_0^n)$, over which the marked points identify with the generic projection points t_i of A_0^n . For a nonzero invariant differential η on A_0 , the pullbacks $t_i^*\eta$ will form a basis of $\Omega_{F'/\kappa}$. These are absolute differentials over κ , not relative differentials over F' : the coordinates of the marked points are themselves parameters in F' . We will bound how much the divisor index and the rank of these forms can decrease after finite extensions whose degree has small p -part.

For context, arbitrary-index genus-one curves over infinite fields finitely generated over their prime fields are constructed by Clark and Lacy [9]. The fields here are instead finitely generated over κ , and our curve must carry the independent marked differential labels as well as the prescribed index. We prove both properties for one translated cyclic twist.

4.1 An elliptic curve with arbitrarily deep torsion

An elliptic curve in characteristic p is ordinary precisely when it has nonzero geometric p -torsion. We construct this property directly.

Lemma 4.1. *For every prime p there is an elliptic curve A_0/κ with a point of order p . On this same curve, for every $s \geq 1$ there is a point $P \in A_0(\kappa)$ of exact order p^s .*

Proof. For $p = 2$, take the Weierstrass cubic

$$y^2 + xy = x^3 + 1.$$

The affine partial derivatives could vanish simultaneously only at $(x, y) = (0, 0)$, which is not on the curve. In its projective equation $Y^2Z + XYZ = X^3 + Z^3$, the origin $[0 : 1 : 0]$ is smooth because the partial derivative with respect to Z is nonzero there. Its points over $\mathbb{F}_2$ are

$$[0 : 1 : 0], \quad (0, 1), \quad (1, 0), \quad (1, 1).$$

Thus its finite group of rational points contains an element of order 2, by Sylow theory.

Suppose that p is odd and put $r = (p - 1)/2$. Consider

$$f_\lambda(x) = x(x - 1)(x - \lambda), \quad H(\lambda) = [x^{p-1}]f_\lambda(x)^r.$$

Here $[x^a]$ means the coefficient of x^a . The coefficient of λ^r in H is $(-1)^r$: in $x^r(x - 1)^r(x - \lambda)^r$, take $(-\lambda)^r$ from the last factor and x^r from the middle factor. In particular H is not the zero polynomial. Choose $\lambda \in \kappa \setminus \{0, 1\}$ with $H(\lambda) \neq 0$. The cubic

$$A_0 : y^2 = f_\lambda(x)$$

is smooth, since f_λ has three distinct roots and the point at infinity is smooth.

Choose $q = p^\ell$ with $\lambda \in \mathbb{F}_q$. For each $x \in \mathbb{F}_q$, the number of solutions for y is $1 + \chi_q(f_\lambda(x))$, where χ_q is the quadratic character with values in $\{0, 1, -1\}$. Its reduction in $\mathbb{F}_q$ is $f_\lambda(x)^{(q-1)/2}$. Reducing the point count modulo p therefore gives

$$\#A_0(\mathbb{F}_q) \equiv 1 + \sum_{x \in \mathbb{F}_q} f_\lambda(x)^{(q-1)/2} = 1 - H(\lambda)^{1+p+\dots+p^{\ell-1}}. \quad (4.1)$$

We verify the last equality, including its coefficient calculation. The exponents occurring in $f_\lambda(x)^{(q-1)/2}$ range from $(q-1)/2$ to $3(q-1)/2$. Among them, only $q-1$ is a positive multiple of $q-1$. Since $\mathbb{F}_q^\times$ is cyclic, the sum of x^a over $\mathbb{F}_q$ is zero for the other positive exponents and is -1 for $a = q-1$. Furthermore,

$$f_\lambda(x)^{(q-1)/2} = \prod_{j=0}^{\ell-1} (f_\lambda(x)^r)^{p^j}.$$

A contribution to the coefficient of x^{q-1} would choose exponents $a_j \in [r, 3r]$ satisfying

$$\sum_{j=0}^{\ell-1} p^j a_j = p^\ell - 1.$$

Reduction modulo p forces $a_0 \equiv p-1 \pmod{p}$. The interval $[r, 3r]$ has length $p-1$ and contains exactly one such integer, namely $p-1$. Subtract this integer and divide the displayed equality by p . Repetition forces $a_j = p-1$ for every j . Thus the desired coefficient is exactly $\prod_j H(\lambda)^{p^j}$, proving (4.1).

The nonzero element $c = H(\lambda)^{1+p+\dots+p^{\ell-1}}$ lies in $\mathbb{F}_p^\times$. Replace $\mathbb{F}_q$ by $\mathbb{F}_{q^d}$, where d is a multiple of the multiplicative order of c . The corresponding expression is $c^d = 1$, so (4.1) shows that $p \mid \#A_0(\mathbb{F}_{q^d})$. Sylow theory again supplies a point of order p .

Finally, multiplication by p is a finite locally free map of degree p^2 on an elliptic curve, by [Foundation 11](#). It is surjective, hence surjective on points over the algebraically closed field κ . If Q has order p^a and $[p]Q' = Q$, then $[p^{a+1}]Q' = 0$ whereas $[p^a]Q' \neq 0$. Starting from a point of order p and repeatedly choosing such a preimage proves the assertion. $\square$

4.2 Cycle sums and rational maps to a constant elliptic curve

The twist below uses $n = p^s$, a point $P \in A_0(\kappa)$ of exact order n , and the field $F' = \kappa(A_0^n)$ with generic projection points t_i . Its field automorphism and semilinear descent action on $A_{0,F'}$ will be

$$\sigma(t_i) = t_{i+1} - P, \quad \delta(x) = \sigma(x) + P \quad (i \text{ modulo } n).$$

The opposite translations make δ permute the marked points. To test the degree b of a divisor on the descended curve, we will pull it back to $A_{0,F'}$ and take its geometric group-law sum v . Invariance of the divisor under δ will give

$$\sigma(v) + [b]P = v.$$

The first lemma shows that v is rational over F' , including when the divisor has inseparable residue fields. Such a point is a rational map $A_0^n \dashrightarrow A_0$. The second lemma writes it as a constant plus one endomorphism in each coordinate. This will let us compare the coordinate endomorphisms in the displayed descent relation and determine which degrees b are possible.

Lemma 4.2. *Let A/κ be an elliptic curve and U/κ a field extension. For every zero-cycle z on A_U , its group-law sum over an algebraic closure of U , using the full geometric multiplicities, belongs to $A(U)$. This sum is additive for signed cycles. Translation of a cycle of degree b by a point $Q \in A(U)$ adds $[b]Q$ to its sum.*

Proof. For a point with inseparable residue field, its multiplicity in the geometric cycle must be included before applying Galois descent to the sum. We first show that multiplication by a power of p moves the point into the corresponding field of p th powers. Write $K_A = \kappa(A)$. A nonzero invariant differential η generates $\Omega_{K_A/\kappa}$, and $[p]^*\eta = 0$ by [Foundation 11](#). Thus every rational function pulled

back by $[p]$ has differential zero. The kernel of $d : K_A \rightarrow \Omega_{K_A/\kappa}$ is exactly K_A^p . Indeed, it is a subfield containing K_A^p , it is a proper subfield because $\Omega_{K_A/\kappa}$ has dimension one, and $[K_A : K_A^p] = p$ by [Foundation 4](#). Consequently

$$[p]^* K_A \subseteq K_A^p. \quad (4.2)$$

For every extension V/κ , this entails

$$[p](A(V)) \subseteq A(V^p). \quad (4.3)$$

Here $\kappa \subseteq V^p$ because κ is perfect. To justify evaluation in (4.3), take $R \in A(V)$ and a κ -affine open W containing $[p]R$. For $f \in \mathcal{O}_A(W)$, write $[p]^* f = g^p$ with $g \in K_A$, using (4.2). On $[p]^{-1}W$ the rational function g is regular: its p th power is regular, and each local ring of this smooth curve is integrally closed. Hence $f([p]R) = g(R)^p \in V^p$ for every such f . The point $[p]R$ therefore factors through $W(V^p)$, as claimed. Iteration gives $[p^u](A(V)) \subseteq A(V^{p^u})$.

It suffices to treat a closed point z with residue field T/U . Let T_{sep} be the maximal separable subextension and write $[T : T_{\text{sep}}] = p^u$. Then $T^{p^u} \subseteq T_{\text{sep}}$. Over an algebraic closure $\bar{U}$, the cycle of z consists of the points R_τ indexed by the U -embeddings $\tau : T_{\text{sep}} \hookrightarrow \bar{U}$, each with multiplicity p^u . To see the multiplicity, first split the separable extension; each remaining purely inseparable tensor factor is local of dimension p^u over $\bar{U}$, with residue field $\bar{U}$.

Each embedding τ extends uniquely to T inside $\bar{U}$. By (4.3),

$$[p^u]R_\tau \in A(\tau(T_{\text{sep}})).$$

The geometric group-law sum is therefore $\sum_\tau [p^u]R_\tau \in A(U^{\text{sep}})$, and Galois permutes its summands. Galois descent for points gives a point of $A(U)$. Additivity proves the assertion for arbitrary signed cycles. Finally, in the geometric sum each copy of each point gains Q under translation, so the change is exactly $[b]Q$. $\square$

Lemma 4.3. *Let A/κ be an elliptic curve.*

- (i) *For every field extension V/κ , an origin-preserving morphism $A_V \rightarrow A_V$ is a group endomorphism defined over κ .*
- (ii) *Every rational map $A^n \dashrightarrow A$ has a unique expression*

$$(x_1, \dots, x_n) \mapsto c + \sum_{i=1}^n \beta_i(x_i), \quad c \in A(\kappa), \quad \beta_i \in \text{End}_\kappa(A). \quad (4.4)$$

In particular, it extends to a morphism on the whole product.

Proof. We first prove the homomorphism assertion in (i). It can be checked after extending V to an algebraic closure. If $f(0) = 0$, form

$$g(x, y) = f(x + y) - f(x) - f(y).$$

Choose a point $a \neq 0$ of the target elliptic curve. The projection to the second factor of $g^{-1}(a)$ is closed, by properness, and does not contain 0, because $g(x, 0) = 0$. Its complement is a nonempty open set W . For $y \in W$, the morphism $x \mapsto g(x, y)$ misses a . A nonconstant morphism of projective integral curves is surjective, so this morphism is constant; its value at $x = 0$ is zero. Thus g is zero on $A \times W$, and hence everywhere. This proves that f is a homomorphism.

We next descend the homomorphism to κ , first by controlling its values on $A(\kappa)$ and then by recovering its rational functions from those values. The zero homomorphism already descends, so

suppose that f is nonconstant. Because $\kappa = \overline{\mathbb{F}}_p$, the curve, its origin, and any chosen point of $A(\kappa)$ are defined over a common finite field. The curve has a finite group of points over that field, so every point of $A(\kappa)$ is torsion. Since f is a homomorphism, each such point has torsion image. For each positive integer N , the scheme $A[N]$ is finite over the algebraically closed field κ ; all its points with values in an extension field are therefore κ -points. Consequently

$$f(A(\kappa)) \subseteq A(\kappa). \quad (4.5)$$

For a rational function $h \in \kappa(A)$, write its pullback as

$$f^*h = \frac{a}{b}, \quad a = \sum_{j=1}^t c_j a_j, \quad b = \sum_{j=1}^t c_j b_j, \quad a_j, b_j \in \kappa(A),$$

where the finitely many $c_j \in V$ are linearly independent over κ . Such an expression follows by putting a rational function in $V(A)$ over a common denominator and choosing a basis for its finitely many scalar coefficients. Fix j_0 with $b_{j_0} \neq 0$. For all but finitely many $R \in A(\kappa)$, all functions in this expression can be evaluated, $b(R)b_{j_0}(R) \neq 0$, and $f^*h(R) \in \kappa$, by (4.5). Writing $\lambda_R = f^*h(R)$, scalar independence gives

$$a_j(R) = \lambda_R b_j(R) \quad \text{for every } j, \quad f^*h(R) = \frac{a_{j_0}(R)}{b_{j_0}(R)}.$$

There are infinitely many such κ -points. A nonzero rational function on a projective integral curve has only finitely many zeros, so $f^*h = a_{j_0}/b_{j_0} \in \kappa(A)$. Apply this to coordinate functions on a κ -affine open of the target. The morphism f is thus a rational map defined over κ . It extends over every point of the smooth source curve by the valuative criterion for properness, and the extension agrees with f after scalar extension. This proves (i).

For (ii), view a rational map in the first variable over $V = \kappa(A^{n-1})$, the function field of the remaining factors. It extends to a morphism $A_V \rightarrow A_V$: the local rings at closed points of the smooth source curve are DVRs, and properness extends the map uniquely over each of them. These extensions are morphisms on neighborhoods and glue by separatedness. Subtracting its value at the origin leaves, by (i), an endomorphism $\beta_1 \in \text{End}_\kappa(A)$. The subtracted value is an element of $A(V)$, hence a rational map from the remaining $n-1$ factors. Repeat to obtain (4.4). The right-hand side is a morphism everywhere. Its value at the origin determines c , and its restriction to each factor with the other coordinates zero determines the corresponding β_i . This also proves uniqueness. $\square$

4.3 The translated cyclic twist and its index

For a regular projective integral curve over a field, its *index* is the positive generator of the subgroup of $\mathbb{Z}$ consisting of divisor degrees; equivalently, it is the greatest common divisor of the degrees of its closed points. Degrees throughout are taken over the stated field.

Proposition 4.4. *Fix $s \geq 1$, put $n = p^s$, and choose A_0/κ and $P \in A_0(\kappa)$ of exact order n as in Lemma 4.1. There are finitely generated fields $F_0 \subset F'$ over κ , each of transcendence degree n , and a smooth projective geometrically integral genus-one curve E/F_0 with the following properties:*

- (i) $F' = \kappa(A_0^n)$, and F'/F_0 is cyclic Galois of degree n . With $t_i \in A_0(F')$ the generic projection points, a generator satisfies

$$\sigma(t_i) = t_{i+1} - P \quad (i \text{ modulo } n). \quad (4.6)$$

- (ii) Under an identification $E_{F'} \simeq A_{0,F'}$, the descent action on points is $x \mapsto \sigma(x) + P$. The two disjoint divisors

$$\mathcal{B}_{F'} = \sum_{i=1}^n [t_i], \quad D_{F'} = \sum_{j=0}^{n-1} [jP] \quad (4.7)$$

descend to finite étale effective divisors $\mathcal{B}, D$ of degree n on E , and D is ample.

- (iii) The index of E/F_0 is exactly n .

Proof. The cyclic permutation followed by translation on the smooth product A_0^n defines the automorphism of its function field in (4.6). Its powers satisfy

$$\sigma^a(t_i) = t_{i+a} - [a]P.$$

Thus $\sigma^n = 1$. If $0 < a < n$, the maps given by the independent projections t_i and t_{i+a} cannot differ by a constant: vary one factor while holding the other fixed. Therefore $\sigma^a \neq 1$. Set $F_0 = (F')^{\langle \sigma \rangle}$. Finite fixed-field theory gives $[F' : F_0] = n$ with the indicated cyclic Galois group. Since F'/κ is finitely generated of transcendence degree n , the same is true of F_0/κ . More explicitly, choose a transcendence basis of F_0/κ ; it is also a transcendence basis of F'/κ , and F' is finite over the rational function field on this basis, as is its intermediate field F_0 .

On $A_{0,F'}$ use the semilinear action

$$\delta(x) = \sigma(x) + P.$$

Since σ fixes P and $[n]P = 0$, this action satisfies $\delta^n = 1$. The translation by $-P$ in the field action cancels the translation by P here, so δ sends t_i to t_{i+1} . It also sends jP to $(j+1)P$. Thus the generic projections give the invariant marked divisor $\mathcal{B}_{F'}$, and the orbit of the origin gives the separate invariant divisor $D_{F'}$ of degree n . All their points are distinct, and the two supports are disjoint: a generic projection is nonconstant, so cannot equal a constant point jP .

We use $D_{F'}$ to descend the curve projectively; once descended, its degree will also give the upper bound n for the index. The invertible sheaf $\mathcal{O}(D_{F'})$ has a canonical linearization, obtained by applying the action to rational functions with prescribed poles in the invariant divisor. It is ample: its third power is the tensor product of the translates of $\mathcal{O}(3[0])$ by the points jP , and $\mathcal{O}(3[0])$ is the ample Weierstrass hyperplane bundle. Thus projective Galois descent, in the form of [Foundation 6](#), constructs E and descends both divisors. Smoothness, geometric integrality, finiteness, and the étale property follow after the splitting extension; ampleness descends as well. Cohomology and base change give $\dim_{F_0} H^1(E, \mathcal{O}_E) = 1$, so E has genus one.

It remains to prove that every divisor degree is divisible by n . The two preparatory lemmas reduce this to comparing endomorphisms in the cyclic descent relation. Let Z be an arbitrary divisor on E of degree b , and pull it back to $A_{0,F'}$. Pullback preserves its degree. By [Lemma 4.2](#), its geometric group-law sum, including inseparable multiplicities and signed coefficients, is a point $v \in A_0(F')$. The pulled-back divisor is invariant under δ . Applying the translation formula in that lemma yields

$$\sigma(v) + [b]P = v. \quad (4.8)$$

By [Lemma 4.3](#), the point v , viewed as a rational map from A_0^n to A_0 , has the form

$$v = c + \sum_{i=1}^n \beta_i(t_i), \quad c \in A_0(\kappa), \quad \beta_i \in \text{End}_\kappa(A_0).$$

Substitution in (4.8) gives

$$\sum_{i=1}^n (\beta_{i-1} - \beta_i)(t_i) + [b]P - \sum_{i=1}^n \beta_i(P) = 0, \quad \beta_0 = \beta_n. \quad (4.9)$$

This equality at the generic point is an equality of morphisms on the product. Evaluate it first at the origin and then on one factor at a time. Each $\beta_{i-1} - \beta_i$ must be zero, so all β_i equal one endomorphism β . Their translation terms sum to

$$\sum_i \beta_i(P) = \beta([n]P) = 0.$$

Equation (4.9) now gives $[b]P = 0$. Since P has exact order n , this proves $n \mid b$. The divisor D has degree n , so the index is exactly n . $\square$

The marked divisor also has the differential property stated at the start of the section. Fix a nonzero invariant differential η on A_0 . The product formula for Kähler differentials, evaluated at the generic point of A_0^n , gives

$$\Omega_{F'/\kappa} = \bigoplus_{i=1}^n F' t_i^* \eta. \quad (4.10)$$

Thus the same splitting field that supplies the marked points makes their absolute differential forms independent.

4.4 Consequences after finite extension

Corollary 4.5. *Let $h \geq 1$ and $s \geq h$, and let F/F_0 be a finite extension with $[F : F_0]_p \leq p^h$. Every divisor on E_F has degree over F divisible by p^{s-h} . More generally, if C/F is a regular projective integral curve with a finite dominant morphism $C \rightarrow E_F$, then every divisor and every invertible sheaf on C has degree over F divisible by p^{s-h} .*

Proof. Put $N = [F : F_0]$. For a closed point z of E_F mapping to x of E , finite pushforward is $z \mapsto [\kappa(z) : \kappa(x)]x$. Its degree over F_0 is

$$[\kappa(z) : \kappa(x)] [\kappa(x) : F_0] = [\kappa(z) : F_0] = N[\kappa(z) : F].$$

Thus a divisor of degree b over F pushes forward to a divisor of degree Nb over F_0 , including when F/F_0 or a point's residue extension is inseparable. By Proposition 4.4, $p^s \mid Nb$. Since $N_p \leq p^h$, it follows that $p^{s-h} \mid b$.

For $C \rightarrow E_F$, the same residue-field tower calculation shows that pushforward preserves the degree over F . Apply the first assertion to this pushforward. Finally every invertible sheaf on a regular integral curve has a nonzero rational section and is the invertible sheaf of its divisor, so the line-bundle assertion follows. $\square$

Lemma 4.6. *Let F/F_0 be a finite extension with $[F : F_0]_p \leq p^h$, and choose a compositum $S = FF'$. Then S/F is finite Galois of degree dividing n , and*

$$[S : F']_p \leq p^h. \quad (4.11)$$

Fix a nonzero invariant differential η on A_0 . For every subset $I \subseteq \{1, \dots, n\}$, the forms $t_i^ \eta$, $i \in I$, have rank at least $|I| - h$ in $\Omega_{S/\kappa}$. The covectors*

$$(1, -t_i^* \eta) \in S \oplus \Omega_{S/\kappa}, \quad i \in I, \quad (4.12)$$

also have rank at least $|I| - h$.

Proof. Since F'/F_0 is Galois, its compositum with any finite extension F/F_0 , including an inseparable one, is Galois over F , with Galois group a subgroup of $\text{Gal}(F'/F_0)$. Consequently $[S : F]$ divides $n = p^s$. The tower identity

$$[S : F'] [F' : F_0] = [S : F] [F : F_0]$$

then gives (4.11).

The forms $t_i^* \eta$ are the basis in (4.10). Their loss of independence over S is measured by the kernel of the first map in the exact sequence

$$S \otimes_{F'} \Omega_{F'/\kappa} \longrightarrow \Omega_{S/\kappa} \longrightarrow \Omega_{S/F'} \longrightarrow 0.$$

The first two spaces both have dimension n over S , since the fields are finitely generated of transcendence degree n over the perfect field κ . The dimension of the kernel of the first map is therefore $\dim_S \Omega_{S/F'}$. Let S_{sep}/F' be the maximal separable subextension and write $[S : S_{\text{sep}}] = p^u$. By (4.11), $u \leq h$. The purely inseparable extension S/S_{sep} can be generated by at most u elements: every nontrivial successive adjunction has degree at least p . Their differentials generate $\Omega_{S/S_{\text{sep}}}$, while $\Omega_{S_{\text{sep}}/F'} = 0$. The differential sequence thus gives $\dim_S \Omega_{S/F'} \leq u \leq h$.

Any subspace spanned by $|I|$ of the original basis vectors loses at most h dimensions under a map with kernel dimension at most h . This proves the assertion for the forms. Projection of (4.12) onto the second summand gives their negatives, so the rank of the covectors is at least the rank of the forms. $\square$

The extra coordinate in (4.12) records the coefficient of the elliptic curve's invariant differential. Indeed, on the constant curve $A_{0,S}$, absolute cotangents split into the line generated by η and the constant differentials from S . Restriction along the marked section t_i is therefore

$$S \oplus \Omega_{S/\kappa} \longrightarrow \Omega_{S/\kappa}, \quad (\lambda, \beta) \longmapsto \lambda t_i^* \eta + \beta.$$

Its kernel is the line generated by $(1, -t_i^* \eta)$. These cotangent directions normal to the marked sections will give the ramification constraints in Section 5.

5 Absolute differentials and the inseparable genus estimate

The two properties of the marked curve in Section 4 will now give an exact lower bound for the arithmetic genus of its purely inseparable covers. Throughout this section, a curve over a field U means a regular integral projective scheme of dimension one over U . It need not be geometrically regular or geometrically reduced. We set

$$\chi(C) = \dim_U H^0(C, \mathcal{O}_C) - \dim_U H^1(C, \mathcal{O}_C), \quad \nu(C) = -2\chi(C).$$

Divisor degrees and cohomology dimensions are always taken over the stated ground field, and the degree of a vector bundle is the degree of its determinant. Thus $\nu(C)$ retains any contribution from an inseparable extension of the constant field.

Fix a finite group J as in Theorem 3.1 and put $m = |J|$. Only this order enters the following estimate, as a bound on the cover degrees needed later. Put $\kappa = \mathbb{F}_p$. For each s , use the fields, elliptic curve, and labels constructed in Proposition 4.4 for that value of s . In particular, $n = p^s$, $F' = \kappa(A_0^n)$, and $E_{F'} \simeq (A_0)_{F'}$. Write $\mathcal{B}$ for the descended finite étale divisor of labels; over F' it is the sum of the distinct rational points $t_1, \dots, t_n$. For $h \geq 1$, $s \geq h$, and a finite extension F/F_0 with $[F : F_0]_p \leq p^h$, we use two properties of the marked curve. Every divisor degree on E_F is divisible by p^{s-h} , by Corollary 4.5. Over $S = FF'$, any set of the forms $t_i^* \eta$ loses at most h dimensions of independence, by Lemma 4.6, where η is a nonzero invariant differential on A_0 .

Theorem 5.1 (Exact purely inseparable genus estimate). *Fix $h \geq 1$ and take $s \geq h$ such that*

$$p^{s-h} > m(h + \log_p m). \quad (5.1)$$

Let F/F_0 be finite with $[F : F_0]_p \leq p^h$, and let Z be the normalization of E_F in a finite purely inseparable extension of its function field of degree $d \leq m$. Choose a compositum $S = FF'$. The curve Z_S is integral and regular. Let r_i be the ramification index of $Z_S \rightarrow E_S$ at the unique point over t_i . With $\nu(Z)$ computed over F , one has

$$\frac{\nu(Z)}{d} \geq \sum_{i=1}^n \left(1 - \frac{1}{r_i}\right). \quad (5.2)$$

The bound is uniform in F , Z , and the tuple-class multiset used later in the counting argument.

The proof has two parts. Absolute differentials give the same inequality with an error of at most $h + \log_p d$. After multiplication by d , the difference between the two sides is a difference of divisor degrees over F , hence a multiple of p^{s-h} . The strict threshold then forces this difference to be nonnegative. We develop the differential identities first, together with the separable ramification bound needed later in [Section 7](#).

5.1 Divisor degrees, duality, and separable ramification

The required identities hold over more general fields than the twisted-curve construction. Throughout this and the next subsection, let κ be any perfect field of characteristic p , let S/κ be finitely generated, and put $\rho = \text{trdeg}_\kappa S$. Curves and degrees in these two subsections are over S .

The classical theory of inseparable genus change begins with Tate [36]; Schröer [33] gives a modern account for geometrically integral curves. Here absolute differentials retain arithmetic genus without a geometric-reducedness hypothesis and also measure ramification at the marked points.

Lemma 5.2 (Degrees over the stated ground field). *For an invertible sheaf $\mathcal{L}$ on C ,*

$$\chi(C, \mathcal{L}) = \deg_S \mathcal{L} + \chi(C). \quad (5.3)$$

A finite dominant morphism $f : C \rightarrow C'$ of such curves is finite flat. If its function-field degree is δ , then

$$\deg_S f^* D = \delta \deg_S D \quad \text{and} \quad \deg_S f_* D' = \deg_S D'$$

for divisors D on C' and D' on C . These assertions, and Euler characteristics, add over disjoint unions of regular integral curves. After a finite separable extension S_1/S , divisor and line-bundle degrees and coherent Euler characteristics have the same numerical values over S_1 as before extension over S .

Proof. For a closed point x put $\deg_S(x) = [\kappa(x) : S]$. For any divisor D the quotient in

$$0 \longrightarrow \mathcal{O}_C(D) \longrightarrow \mathcal{O}_C(D+x) \longrightarrow \mathcal{Q}_x \longrightarrow 0$$

has length one over the DVR $\mathcal{O}_{C,x}$, so its dimension over S is $[\kappa(x) : S]$. Finite-support sheaves have no higher cohomology. Adding and subtracting closed points proves $\chi(\mathcal{O}_C(D)) = \chi(C) + \deg_S D$. A rational trivialization of $\mathcal{L}$ identifies it with $\mathcal{O}_C(D)$ for a divisor D , giving (5.3). Applied to a principal divisor, the same formula gives degree zero, so this degree depends only on $\mathcal{L}$.

Locally on C' , the finite module $f_*\mathcal{O}_C$ is torsion-free over a DVR and therefore free, of generic rank δ . For a closed point $x \in C'$ and a uniformizer t there, reduction of this free module modulo t gives

$$\sum_{y \mapsto x} e_y [\kappa(y) : \kappa(x)] = \delta, \quad (5.4)$$

where e_y is the valuation of t in $\mathcal{O}_{C,y}$. Indeed the summand at y has a filtration of length e_y with successive quotients $\kappa(y)$. Multiplication by $[\kappa(x) : S]$ proves the pullback degree formula. The pushforward formula follows from $f_*y = [\kappa(y) : \kappa(x)]x$ and the tower formula for residue degrees.

Finite separable scalar extension preserves regularity, and a regular curve after such extension is a disjoint union of its integral components. Flat base change for coherent cohomology in [Foundation 9](#) gives

$$H^i(C_{S_1}, \mathcal{G}_{S_1}) = H^i(C, \mathcal{G}) \otimes_S S_1.$$

Thus the dimensions over the respective fields, and hence Euler characteristics, agree. Applying (5.3) proves the assertion for line bundles and their Cartier divisors. In particular it also applies to a reduced finite divisor: separable scalar extension preserves its reducedness. $\square$

Proposition 5.3 (The absolute determinant computes arithmetic genus). *The absolute cotangent sheaf $\Omega_{C/\kappa}$ is locally free of rank $\rho + 1$, and*

$$\deg_S \det \Omega_{C/\kappa} = \nu(C). \quad (5.5)$$

Proof. An affine coordinate ring of C is a localization of a finitely generated integral κ -algebra: choose a finitely generated domain with fraction field S , clear the finitely many coefficients in a presentation over S , and take the contraction of the defining prime ideal before localizing. Its local rings are consequently essentially of finite type over κ . The regularity criterion over a perfect field in [Foundation 8](#) says that these regular local rings are local rings at smooth points of the corresponding κ -variety. Their absolute differentials are free of rank

$$\mathrm{trdeg}_\kappa \kappa(C) = \rho + 1.$$

The sheaf is coherent: the relative differentials over S are coherent, and $\Omega_{S/\kappa}$ is finite-dimensional, so the cotangent sequence gives a finite presentation locally. This proves the asserted local freeness without making a smoothness assumption over S .

Choose a projective embedding $i : C \hookrightarrow \mathbb{P}_S^b$ and let $\mathcal{I}$ be its ideal sheaf. This is a regular immersion of codimension $b - 1$. Here is a local verification. Let R be an ambient regular local ring, I the local ideal, and R/I the regular local ring of C . Their dimensions differ by $b - 1$, by the dimension formula at the common residue field. Since both are regular, the kernel of

$$\mathfrak{m}_R / \mathfrak{m}_R^2 \longrightarrow \mathfrak{m}_{R/I} / \mathfrak{m}_{R/I}^2$$

has dimension $b - 1$. Choose $b - 1$ elements of I lifting a basis of this kernel. They form part of a regular system of parameters of R and cut out a regular local quotient of dimension $\dim(R/I)$. The remaining ideal defining R/I in that quotient is prime and must be zero: a nonzero prime in a finite-dimensional local domain strictly lowers the quotient dimension. Thus the chosen elements generate I . It follows that $\mathcal{I}/\mathcal{I}^2$ is locally free of rank $b - 1$.

The absolute conormal sequence is in fact short exact:

$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow i^* \Omega_{\mathbb{P}_S^b/\kappa} \longrightarrow \Omega_{C/\kappa} \longrightarrow 0. \quad (5.6)$$

To check the left end, the kernel of the last arrow is locally free, since the last module is locally free. Its rank is $(b + \rho) - (1 + \rho) = b - 1$. The conormal sequence surjects $\mathcal{I}/\mathcal{I}^2$ onto this kernel. A surjection between locally free modules of the same rank is an isomorphism, proving (5.6).

The projective space is obtained from $\mathbb{P}_\kappa^b$ by extending constants. Its absolute cotangents therefore split into relative cotangents over S and the constant bundle of $\Omega_{S/\kappa}$. Taking determinants in (5.6) gives

$$\det \Omega_{C/\kappa} \simeq \mathcal{L}_c \otimes_S \det \Omega_{S/\kappa}, \quad \mathcal{L}_c = \mathcal{O}_C(-b-1) \otimes \det(\mathcal{I}/\mathcal{I}^2)^\vee.$$

The last tensor factor is a constant line and has degree zero.

We compute the degree of $\mathcal{L}_c$ using the projective-space duality in [Foundation 10](#). A Koszul resolution for the local regular sequence defining C shows that

$$\mathcal{E}xt_{\mathbb{P}_S^b}^v(i_*\mathcal{O}_C, \mathcal{O}(-b-1)) = \begin{cases} i_*\mathcal{L}_c, & v = b-1, \\ 0, & v \neq b-1. \end{cases}$$

After dualizing the Koszul resolution, its top cohomology is the dual of the top exterior power of the free module of equations, restricted to C . A change of those equations acts on this exterior power by the determinant of their change on $\mathcal{I}/\mathcal{I}^2$. Thus the local top cohomology modules glue to $\det(\mathcal{I}/\mathcal{I}^2)^\vee \otimes \mathcal{O}_C(-b-1)$.

The local-to-global Ext spectral sequence has only this one nonzero row. Projective-space duality consequently gives, for $r = 0, 1$,

$$H^r(C, \mathcal{O}_C)^\vee \simeq \mathcal{E}xt_{\mathbb{P}_S^b}^{b-r}(i_*\mathcal{O}_C, \mathcal{O}(-b-1)) \simeq H^{1-r}(C, \mathcal{L}_c).$$

Hence $\chi(C, \mathcal{L}_c) = -\chi(C)$. Applying (5.3) yields $\deg_S \mathcal{L}_c = -2\chi(C)$, which proves (5.5). $\square$

Lemma 5.4 (A separable determinant bound, including the wild term). *Let $f : C \rightarrow C'$ be a finite dominant, generically separable morphism of regular integral projective curves over S , of degree δ . There is an effective divisor R_f on C such that*

$$\nu(C) = \delta\nu(C') + \deg_S R_f.$$

If $y \in C$ has ramification index a_y over $f(y)$, then

$$\text{ord}_y R_f \geq a_y - 1 + \mathbf{1}_{p|a_y}. \quad (5.7)$$

No separability of $\kappa(y)/\kappa(f(y))$ is required. The formula and the inequalities can be added componentwise after a finite separable extension of S .

Proof. The morphism $f^*\Omega_{C'/\kappa} \rightarrow \Omega_{C/\kappa}$ is generically an isomorphism: at the function fields a finite separable extension has no relative differentials, and both absolute differential spaces have dimension $\rho + 1$. Its determinant is therefore a nonzero regular morphism of line bundles. The divisor of this morphism is effective; call it R_f . Its degree is the difference of the two determinant degrees, which is $\nu(C) - \delta\nu(C')$ by [Lemma 5.2](#) and [Proposition 5.3](#).

Write $x = f(y)$ and choose a uniformizer t of the DVR $B = \mathcal{O}_{C',x}$. Its absolute differential is primitive in $\Omega_{B/\kappa}$, meaning that it is part of a B -basis. Indeed the conormal sequence for the residue field is

$$(t)/(t^2) \longrightarrow \Omega_{B/\kappa} \otimes_B \kappa(x) \longrightarrow \Omega_{\kappa(x)/\kappa} \longrightarrow 0.$$

The middle term has dimension $\rho + 1$ and the right term dimension ρ , since $\kappa(x)$ is finite over S and [Foundation 4](#) computes absolute differential dimension over the perfect field κ . Thus the first

map has nonzero one-dimensional image, generated by dt . Consequently dt is nonzero modulo the maximal ideal, and it extends to a basis of the free module $\Omega_{B/\kappa}$.

In $A = \mathcal{O}_{C,y}$ write $t = u\tau^{a_y}$, where u is a unit and τ a uniformizer. Absolute differentiation gives

$$dt = \tau^{a_y} du + a_y u \tau^{a_y-1} d\tau.$$

All coordinates of this column of the differential matrix are divisible by τ^{a_y-1} ; when $p \mid a_y$, the second term is zero, so they are divisible by τ^{a_y} . Expanding the determinant in this column proves (5.7). Every step uses absolute differentials over κ , so an inseparable extension of residue fields causes no change to the argument. $\square$

5.2 The differential calculation for a height-one map

We have established the determinant formula for arithmetic genus and the separable ramification bound. The remaining ingredient is an exact genus identity for a purely inseparable map of exponent one; successive such maps will handle arbitrary purely inseparable covers.

Write $T_C = (\Omega_{C/\kappa})^\vee$ for the absolute tangent bundle. It has rank $\rho + 1$, even when the relative tangent sheaf over S has different behavior.

The calculation below is a curve-level form of the canonical-bundle formula for a p -closed foliation; compare [12, Section I.1, Proposition 1.1 and Equations (1.2)–(1.3)]. We prove the version needed here directly from local presentations, using absolute differentials over the perfect constants. In this way the kernel, the Frobenius-power cokernel, and all inseparable residue extensions remain explicit.

Lemma 5.5 (Height-one kernels and cokernels). *Let $\pi : C \rightarrow C'$ be a finite dominant morphism of regular integral projective curves over S such that $\kappa(C)^p \subseteq \kappa(C')$. Put $q = [\kappa(C) : \kappa(C')] = p^a$ and*

$$\mathcal{F} = \ker(T_C \longrightarrow \pi^* T_{C'}), \quad \mathcal{Q} = \operatorname{coker}(T_C \longrightarrow \pi^* T_{C'}).$$

Both $\mathcal{F}$ and $\mathcal{Q}$ are locally free of rank a , and

$$\det \mathcal{Q} \simeq (\det \mathcal{F})^{\otimes p}. \tag{5.8}$$

Consequently

$$\nu(C) = q\nu(C') + (p-1) \deg_S \mathcal{F}. \tag{5.9}$$

Proof. The integral presentation. On an affine open of C' , every element of the finite upstairs coordinate ring has its p th power in the downstairs ring: the power lies in the downstairs fraction field and is integral, so normality puts it in that ring. There is therefore a unique prime upstairs over each downstairs prime, since membership of an element is determined by membership of its p th power. At a closed point we may consequently write $B \subset A$ for the downstairs and upstairs DVRs, with A still finite over B . It is finite free of rank q and satisfies $A^p \subset B$.

Let τ be an upstairs uniformizer, let ϵ be the ramification index, and put $f = [\kappa(A) : \kappa(B)]$. The element $\tau^p \in B$ has upstairs valuation p ; all nonzero elements of B have upstairs valuation in $\epsilon\mathbb{Z}$. Thus $\epsilon \mid p$, so ϵ is 1 or p . Reduction of the rank- q free module modulo a downstairs uniformizer gives $q = \epsilon f$ by (5.4). The residue extension has exponent at most one. Successively adjoining a generator outside the preceding residue field gives a degree- p extension each time. Choose residue generators $\bar{b}_1, \dots, \bar{b}_r$ in this manner, so their monomials with exponents in $\{0, \dots, p-1\}$ form a $\kappa(B)$ -basis of $\kappa(A)$ and $f = p^r$. Lift them to $b_1, \dots, b_r \in A$. If $\epsilon = p$, append $b_{r+1} = \tau$; if $\epsilon = 1$, append nothing. In both cases there are exactly a generators, and $c_\ell = b_\ell^p$ belongs to B .

For a downstairs uniformizer t , the ideal tA is $\tau^\epsilon A$. Filter A/tA by the powers of τ from 0 to ϵ . Each successive quotient is $\kappa(A)$, and the lifted residue monomials give its basis after multiplication by the corresponding power of τ . It follows that the q monomials

$$b_1^{i_1} \cdots b_a^{i_a}, \quad 0 \leq i_\ell < p,$$

are a $\kappa(B)$ -basis of A/tA . By Nakayama they generate A over B ; since A is free of rank q , they are a B -basis. The homomorphism from the monic-relation algebra therefore gives the actual presentation

$$A \simeq B[X_1, \dots, X_a]/(X_1^p - c_1, \dots, X_a^p - c_a), \quad X_\ell \mapsto b_\ell. \quad (5.10)$$

Both sides have the same free monomial basis over B , which verifies that there are no further relations. This includes the cases of a nontrivial inseparable residue extension and simultaneous residue extension and ramification.

These presentations are available on open neighborhoods. Shrink an open downstairs around the chosen point so that all the b_ℓ are sections of $\pi_*\mathcal{O}_C$, all the c_ℓ are regular downstairs, and the determinant of the displayed monomial basis is invertible. Then (5.10) holds there. Such neighborhoods give local presentations throughout the curves.

Free kernel and free cokernel. Set $N = A \otimes_B B/\kappa$, and let $W \subset N$ be the span of $dc_1, \dots, dc_a$. Differentiating the displayed presentation, using characteristic p , gives

$$\Omega_{A/\kappa} \simeq (N/W) \oplus \bigoplus_{\ell=1}^a A db_\ell. \quad (5.11)$$

Both N and $\Omega_{A/\kappa}$ are free of rank $\rho + 1$ by Proposition 5.3. The summand N/W in (5.11) is therefore free, of rank $\rho + 1 - a$. The sequence $0 \rightarrow W \rightarrow N \rightarrow N/W \rightarrow 0$ splits, so W is free of rank a . Its a generators dc_ℓ are consequently a basis.

Dualizing now shows that the tangent kernel consists of the κ -derivations $A \rightarrow A$ killing B , and has free coordinates

$$D \mapsto (D(b_1), \dots, D(b_a)).$$

The tangent cokernel is $W^\vee$, with coordinates

$$[\delta] \mapsto (\delta(c_1), \dots, \delta(c_a)), \quad \delta \in \text{Der}_\kappa(B, A).$$

Indeed a derivation $B \rightarrow A$ extends to A exactly when it kills the c_ℓ ; when it does, the values assigned to the b_ℓ are arbitrary. Thus both kernel and cokernel are locally free of rank a .

Gluing the determinant lines. On an overlap, write another system of generators as $b'_k = P_k(b_1, \dots, b_a)$ with coefficients in B , and put

$$J_{k\ell} = \frac{\partial P_k}{\partial X_\ell}(b_1, \dots, b_a).$$

A derivation in the kernel kills the coefficients, so its new coordinate vector is J times its old coordinate vector. The matrix J is invertible since both coordinate systems are bases. Taking p th powers of the generator identities gives downstairs

$$c'_k = (b'_k)^p = P_k^{[p]}(c_1, \dots, c_a),$$

where $P_k^{[p]}$ means that the coefficients of P_k are raised to the p th power. Their absolute differentials vanish. Differentiating this identity by a representative $\delta : B \rightarrow A$ of a cokernel class therefore gives

$$\delta(c'_k) = \sum_\ell J_{k\ell}^p \delta(c_\ell).$$

Thus the cokernel coordinate transition is the entrywise Frobenius power $J^{[p]}$. Coordinate transitions for determinant lines are $\det J$ and $\det(J^{[p]}) = (\det J)^p$; if one uses local frames instead, both transitions are inverted. In either convention they give exactly (5.8), with the exponent $+p$.

Finally the exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow T_C \longrightarrow \pi^* T_{C'} \longrightarrow \mathcal{Q} \longrightarrow 0$$

and (5.8) give

$$\deg T_C = q \deg T_{C'} - (p-1) \deg \mathcal{F}.$$

Since $\deg T_C = -\nu(C)$, this is (5.9). $\square$

5.3 Proof of the exact estimate

Return to the fields and marked curve in Theorem 5.1, with $\kappa = \overline{\mathbb{F}}_p$ and $S = FF'$. The height-one identity will be applied along a Frobenius tower from Z_S to E_S . At each stage the label covectors bound the degree of the tangent-map image; the resulting weighted sum gives the bounded-error estimate before the final index argument removes the error.

Proof of Theorem 5.1. Normalization and the separable splitting field. Normalizations are finite by Foundation 7, and normal curves over fields are regular. We first record why the finite purely inseparable maps in this argument are radicial, that is, universally injective. If A is the integral closure of a normal affine domain B in a finite purely inseparable extension, there is an exponent v such that $a^{p^v} \in B$ for every $a \in A$. The power belongs to $\text{Frac}(B)$ and is integral over B , so it belongs to B by normality. After tensoring with any B -algebra, the p^v th power of every element still lies in the image of that algebra. Consequently over a prime of the base there is at most one prime upstairs: membership of an element in a prime is determined by membership of its p^v th power downstairs. The residue extension is purely inseparable for the same reason. This proves the assertion after every base change.

The extension S/F is finite separable, since F'/F_0 is finite Galois. Thus Z_S is regular and reduced by Foundation 8. Its map to the integral curve E_S is still finite, surjective, and radicial. This map is a homeomorphism on underlying spaces, so Z_S is irreducible and hence integral. It is the normalization of E_S in its function field: it is finite and normal with that field, and an element integral over the base is also integral over its finite normal coordinate ring. Its degree remains d by finite flatness. By Lemma 5.2,

$$\nu(Z_S) = \nu(Z),$$

where the two Euler characteristics are computed over their stated fields. We may therefore do the differential calculation over S . If $d = 1$, this curve is E_S , every $r_i = 1$, and the theorem holds. Assume henceforth that $d > 1$.

The Frobenius tower and its recurrence. Let $M_e = S(E)$ and $M_z = S(Z)$, and set

$$M_j = M_e M_z^{p^j} \quad (0 \leq j \leq b), \quad M_0 = M_z, \quad M_b = M_e,$$

where b is the first index for which the last equality holds. Let Y_j be the normalization of E_S in M_j , and write

$$\gamma_j : Y_j \longrightarrow E_S, \quad d_j = [M_j : M_e], \quad \pi_j : Y_j \longrightarrow Y_{j+1}, \quad q_j = d_j/d_{j+1} = p^{m_j}.$$

In particular $Y_0 = Z_S$, $d_0 = d$, $Y_b = E_S$, and $d_b = 1$. The defining fields satisfy

$$M_{j+1} = M_e M_j^p. \tag{5.12}$$

Thus each π_j has exponent at most one and Lemma 5.5 applies.

Put $T_j = (\Omega_{Y_j/\kappa})^\vee$ and

$$\mathcal{F}_j = \ker(T_j \longrightarrow \pi_j^* T_{j+1}).$$

This is also the kernel of the composite tangent map $T_j \rightarrow \gamma_j^* T_{E_S}$. To see this, at the function field a κ -derivation kills M_e if and only if it kills $M_e M_j^p = M_{j+1}$, since every derivation kills p th powers. The two kernels are therefore equal generically. They are equal as sheaves: a section whose image vanishes generically has zero image everywhere in either of the locally free, hence torsion-free, targets.

The genus identity involves $\deg_S \mathcal{F}_j$. We will estimate this degree through the image of the tangent map to E_S . Choose a nonzero invariant differential η on A_0 . The constant-curve identification of E_S gives the splitting

$$\Omega_{E_S/\kappa} \simeq \mathcal{O}_{E_S} \eta \oplus (\mathcal{O}_{E_S} \otimes_S \Omega_{S/\kappa}).$$

Thus its dual is a trivial bundle of rank $n+1$. In this trivial target we can use the marked points to impose linear vanishing conditions on the tangent-map image.

Let $\mathcal{H}_j$ be the image in $\gamma_j^* T_{E_S}$, and put $b_j^\circ = -\deg_S \mathcal{H}_j$. It is a torsion-free coherent sheaf on a regular curve, hence locally free. Its rank is $n+1-m_j$, since $\text{trdeg}_\kappa S = n$. The exact sequence with kernel $\mathcal{F}_j$ and image $\mathcal{H}_j$, together with (5.9), gives

$$\nu(Y_j) = q_j \nu(Y_{j+1}) + (p-1) \deg_S \mathcal{F}_j, \quad b_j^\circ = \nu(Y_j) + \deg_S \mathcal{F}_j.$$

Eliminating $\deg_S \mathcal{F}_j$ and using $d_j = q_j d_{j+1}$ yields

$$\frac{\nu(Y_j)}{d_j} = \frac{1}{p} \frac{\nu(Y_{j+1})}{d_{j+1}} + \left(1 - \frac{1}{p}\right) \frac{b_j^\circ}{d_j}. \quad (5.13)$$

The terminal term $\nu(Y_b)$ is zero because E_S has genus one. It remains to give a lower bound for $b_j^\circ/d_j = -\deg_S \mathcal{H}_j/d_j$.

Vanishing at a ramified label. For each j let S_j be the set of labels at which γ_j has ramification index greater than one. For $i \in S_j$ write $D_{j,i} = \gamma_j^*(t_i)$ and

$$v_i = (1, -t_i^* \eta) \in S \oplus \Omega_{S/\kappa}.$$

Regard v_i as a constant linear functional on $\gamma_j^* T_{E_S}$. We claim that its restriction to $\mathcal{H}_j$ vanishes along the entire divisor $D_{j,i}$; equivalently, its values lie in $\mathcal{O}_{Y_j}(-D_{j,i})$.

Let a be a local parameter defining the rational point t_i on E_S . Restriction of absolute differentials along t_i sends (λ, β) in the displayed splitting to $\lambda t_i^* \eta + \beta$. Its kernel is the line spanned by v_i . Since a restricts to zero, da belongs to this kernel modulo (a) . Moreover its relative component is nonzero: t_i is an S -rational point on a smooth curve over S , and a generates its relative cotangent space. Hence

$$da \equiv \lambda_i v_i \pmod{a \Omega_{E_S/\kappa}} \quad \text{for some } \lambda_i \in S^\times.$$

At the unique point upstairs write $a = u\tau^r$, with u a unit. The nontrivial purely inseparable index r is divisible by p ; therefore

$$d(\gamma_j^* a) = \tau^r du = (\gamma_j^* a) u^{-1} du.$$

It lies in the ideal of the *full* divisor $D_{j,i}$ times $\Omega_{Y_j/\kappa}$. Pulling back the preceding congruence and dividing by the unit λ_i shows that the pullback of v_i lies in that same ideal times $\Omega_{Y_j/\kappa}$. To relate this differential to the claimed functional, denote the tangent map by $d\gamma_j : T_j \rightarrow \gamma_j^* T_{E_S}$ and write $\gamma_j^* v_i$ for the image of the pulled-back covector in $\Omega_{Y_j/\kappa}$. For a local section θ of T_j , duality gives

$$v_i(d\gamma_j(\theta)) = (\gamma_j^* v_i)(\theta).$$

The divisibility just proved makes this value a section of $\mathcal{O}_{Y_j}(-D_{j,i})$. Since $\mathcal{H}_j$ is the image of $d\gamma_j$, the claim follows. The vanishing is along $\gamma_j^*(t_i)$ with its full multiplicity, not just its reduced support.

From independent vanishing conditions to a degree bound. By Lemma 4.6, among the v_i for $i \in S_j$ there are $c \geq |S_j| - h$ independent covectors over S .

Complete the c chosen covectors to a constant basis of the dual of the target trivial tangent bundle. The full-divisor constraint gives an injection

$$\mathcal{H}_j \hookrightarrow \mathcal{O}_{Y_j}^{n+1-c} \oplus \bigoplus_{i \text{ chosen}} \mathcal{O}_{Y_j}(-D_{j,i}). \quad (5.14)$$

Let $R = n + 1 - m_j$ be the rank of $\mathcal{H}_j$. The R th exterior power of the bundle on the right is a direct sum of line bundles, each the tensor product of a choice of R summands. The induced map from $\det \mathcal{H}_j$ is nonzero, so it has a nonzero component in at least one of these lines. Only $n + 1 - c$ of the available summands are trivial. The chosen line must therefore contain at least $R - (n + 1 - c) = c - m_j$ negative divisor summands when this number is positive, and at least zero otherwise. Each $D_{j,i}$ has degree d_j by finite flatness and rationality of t_i . If the nonzero component uses k negative summands, its target line has degree $-kd_j$. A nonzero regular map from one line bundle to another has an effective zero divisor, and consequently

$$\deg_S \mathcal{H}_j \leq -kd_j, \quad \frac{b_j^\circ}{d_j} \geq c - m_j \geq |S_j| - h - m_j.$$

The argument also covers rank zero, with $\det(0) = \mathcal{O}$.

The weighted estimate. Write $r_i = p^{u_i}$. Every height-one step has local index at most p , as proved in Lemma 5.5. The composite $Y_0 \rightarrow Y_j$ consequently has ramification index at most p^j at this label. In particular, $u_i \leq b$. Since ramification indices multiply, the remaining map $Y_j \rightarrow E_S$ has index at least p^{u_i-j} whenever $j < u_i$. Hence $i \in S_j$ for every $0 \leq j < u_i$. Iterating (5.13) with terminal value zero gives

$$\frac{\nu(Z)}{d} = \sum_{j=0}^{b-1} p^{-j} \left(1 - \frac{1}{p}\right) \frac{b_j^\circ}{d_j}.$$

Write $w_j = p^{-j}(1 - 1/p)$ for the weight of stage j . For each label, the weights from those first u_i stages sum to

$$\sum_{j=0}^{u_i-1} p^{-j} \left(1 - \frac{1}{p}\right) = 1 - p^{-u_i} = 1 - \frac{1}{r_i}.$$

Summing these contributions by label gives

$$\sum_{j=0}^{b-1} w_j |S_j| = \sum_{i=1}^n \sum_{\substack{0 \leq j < b \\ i \in S_j}} w_j \geq \sum_{i=1}^n \left(1 - \frac{1}{r_i}\right).$$

The error terms in the degree bound contribute separately. Since $m_j \leq \log_p d$ and the sum of all the weights is $1 - p^{-b} < 1$,

$$\sum_{j=0}^{b-1} w_j (h + m_j) \leq h + \log_p d.$$

Substitute $b_j^\circ/d_j \geq |S_j| - h - m_j$ in the recurrence sum. Subtracting the error bound from the label contribution yields

$$\frac{\nu(Z)}{d} \geq \sum_{i=1}^n \left(1 - \frac{1}{r_i}\right) - (h + \log_p d). \quad (5.15)$$

Divisors over F and exact rounding. Multiplying (5.15) by d bounds the possible deficit by $d(h + \log_p d)$. Set $N = p^{s-h}$ and consider the difference

$$A = \nu(Z) - d \sum_i \left(1 - \frac{1}{r_i}\right).$$

We will realize both terms as degrees of divisors on Z , each retaining its degree under pushforward to E_F . The index property Corollary 4.5 then makes both degrees divisible by N . These divisors must be defined over F , before the separable extension to S . By Proposition 5.3, the integer $\nu(Z)$ is the degree over F of $\det \Omega_{Z/\kappa}$. A rational trivialization represents this line bundle by a Cartier divisor on the regular integral curve Z . Its pushforward to E_F preserves degree over F , by Lemma 5.2. Thus

$$\nu(Z) \in N\mathbb{Z}.$$

This uses the absolute determinant on Z/F and its Euler characteristic over F , including any inseparable constant field.

Let T be the reduced inverse image in Z of $\mathcal{B}_F$. It is an effective divisor on the regular curve Z . Finite separable scalar extension preserves reducedness, so T_S is the reduced inverse image of the split labels. If z_i is the unique point above the rational point t_i , the local degree formula is

$$r_i[\kappa(z_i) : S] = d.$$

It follows that

$$\deg_F T = \deg_S T_S = \sum_i \frac{d}{r_i}.$$

In particular this sum counts the full residue degrees of the reduced points; those residue fields may be inseparable over S . The divisor $\gamma^* \mathcal{B}_F - T$ is defined on Z over F , where $\gamma : Z \rightarrow E_F$, and its degree is

$$dn - \sum_i \frac{d}{r_i} = d \sum_i \left(1 - \frac{1}{r_i}\right).$$

Push it to E_F to conclude that this integer also belongs to $N\mathbb{Z}$.

Consequently A is a multiple of N . By (5.15) and $d \leq m$,

$$A \geq -d(h + \log_p d) \geq -m(h + \log_p m) > -N.$$

A multiple of N strictly greater than $-N$ is nonnegative. This proves (5.2). The sufficient threshold (5.1) contains no datum from F , Z , or a tuple-class multiset. If a threshold independent of the prime is desired, the stronger condition

$$2^{s-h} > m(h + \log_2 m)$$

suffices for every $p \geq 2$. This proves the stated uniformity. $\square$

6 A valued lift and descent of marked covers

Fix the group J , the integer $h \geq 1$, and $n = p^s$ from Theorem 3.1, and retain the characteristic- p data $A_0, P, F', F_0, E, t_1, \dots, t_n$ of Proposition 4.4. Thus P has exact order n , $F' = \kappa(A_0^n)$, and the generator of $\text{Gal}(F'/F_0)$ satisfies $\sigma(t_i) = t_{i+1} - P$, with cyclic indices. All the valued fields constructed in this section are auxiliary; they are independent of the block coefficient system $(K, \mathcal{O}, k)$.

The output sought here is a cover over a field of small p -part degree whenever the finite marked-cover set has a small Sylow orbit. We first lift the curve and its labels, then choose a valued base on which full field degrees survive as residue degrees. This prepares the genus comparison in [Section 7](#) without yet constructing a model of the cover.

6.1 Lifting the twist and its marked points

Proposition 6.1. *There are complete discretely valued fields $K_b \subset K'_b$ of characteristic zero, with valuation rings $R_b \subset R'_b$, and a smooth projective genus-one curve $\mathcal{X}/R_b$ with the following properties.*

- (i) *The extension K'_b/K_b is cyclic Galois of degree n ; the extension R'_b/R_b is finite étale Galois, its residue extension is F'/F_0 , and the two fields have the same value group. Moreover, K_b contains $\mathbb{Q}_p$ and all m th roots of unity, where $m = |J|$.*
- (ii) *The special fiber of $\mathcal{X}$ is E/F_0 . There is a relatively ample effective divisor D on $\mathcal{X}$ of fiber degree n .*
- (iii) *Over R'_b there are pairwise disjoint sections $\xi_1, \dots, \xi_n$ reducing respectively to $t_1, \dots, t_n$. They avoid D , and their union descends to a finite étale effective divisor on $\mathcal{X}$. Under the cyclic descent action the sections are permuted by $\xi_i \mapsto \xi_{i+1}$.*

Proof. We first construct the constant complete DVR. Starting with $\mathbb{Z}_p$, successively lift monic separable irreducible polynomials over the residue fields so that the resulting finite residue fields exhaust $\kappa = \overline{\mathbb{F}}_p$. To justify this construction, if R is the current complete DVR and $f \in R[T]$ is such a monic lift, then $R[T]/(f)$ is finite free, its reduction is a field, and its maximal ideal is generated by the original uniformizer. The polynomial f is irreducible over the fraction field: a monic factorization there would have coefficients integral over R , hence in R , and would give a factorization of its irreducible reduction. Consequently the quotient is a local domain with principal maximal ideal, hence a DVR; its residue extension is the prescribed one. Its derivative is a unit, so the extension is unramified. The union of this tower is a valuation ring with value group $\mathbb{Z}$ and residue κ . Complete it. Completion preserves its uniformizer and residue field, giving a complete DVR of mixed characteristic with residue κ . Denote it for the moment by $R_c^{(0)}$, with fraction field $K_c^{(0)}$.

Lift the coefficients of the smooth Weierstrass equation for A_0 to this DVR. Its discriminant remains a unit, so it defines a smooth elliptic scheme $\mathcal{A}$ by [Foundation 11](#). We next lift P without assuming that the torsion scheme is étale. The kernel $\mathcal{A}[n]$ is finite locally free of degree n^2 . Its generic fiber is dense: in its finite flat coordinate algebra the uniformizer is a nonzerodivisor, so no irreducible component is supported in the special fiber. Since this dense generic fiber is a finite set, one of its points Q has closure containing P . Its field of definition K_Q is finite over $K_c^{(0)}$; this is a characteristic-zero field, not a valuation residue field. The reduced closure of Q is finite and integral over $R_c^{(0)}$. Pass to the integral closure of $R_c^{(0)}$ in K_Q . By [Foundation 5](#) it is a complete DVR, and lying over shows that its closed point maps to P . We obtain an n -torsion section $\tilde{P}$ specializing to P . Its order is exactly n , since its order divides n and reduction has order n . Enlarge the constant field once more to contain all m th roots of unity. These are finite extensions of complete discretely valued fields, and their residue fields are still κ , which is algebraically closed. Denote the resulting constant valuation ring by R_c and choose its uniformizer ϖ .

Consider the smooth product $\mathcal{A}^n/R_c$. Its generic fiber is integral, and flatness makes the total space integral. At the generic point of the integral special fiber, its local ring is a noetherian local domain whose maximal ideal is generated by ϖ and whose residue field is F' . It is therefore a DVR.

Complete its fraction field for this valuation and call the result K'_b , with valuation ring R'_b . The product projections give points $\xi_i \in \mathcal{A}(K'_b)$. Their sections over R'_b reduce to the generic projections t_i .

The automorphism of the product specified by

$$\sigma(\xi_i) = \xi_{i+1} - \tilde{P}$$

fixes the constants, has order n , and preserves the special fiber. It preserves the valuation and extends to K'_b . Its residue action is the specified action on F' , of order n . Put $K_b = (K'_b)^{\langle \sigma \rangle}$. The fixed field is closed in the complete valued field K'_b , so it is complete. The fixed uniformizer ϖ shows that its value group is the same discrete group. Finite fixed-field theory gives $[K'_b : K_b] = n$ with Galois group $\langle \sigma \rangle$.

Let k_b be the residue field of K_b . Then $k_b \subseteq F_0$. Lifting a linearly independent system in a residue extension to units gives a linearly independent system over the valued base field: normalize a putative relation by a coefficient of minimum value and reduce. Applying this observation here gives

$$n = [F' : F_0] \leq [F' : k_b] \leq [K'_b : K_b] = n.$$

Thus $k_b = F_0$.

To verify the stronger assertion about valuation rings, choose a primitive element $\bar{a}$ of the separable extension F'/F_0 and a lift $a \in R'_b$. The polynomial

$$f(T) = \prod_{j=0}^{n-1} (T - \sigma^j a) \in R_b[T]$$

reduces to the separable irreducible polynomial of $\bar{a}$. Consequently $R_b[a]$ is finite free of rank n , local with maximal ideal (ϖ) , and a domain with fraction field K'_b . It is a DVR, and the derivative $f'(a)$ is a unit. Uniqueness of the extension of the complete-DVR valuation identifies it with R'_b . This proves that R'_b/R_b is finite *étale*. The Galois torsor map

$$R'_b \otimes_{R_b} R'_b \longrightarrow \prod_{j=0}^{n-1} R'_b, \quad a \otimes b \longmapsto (a\sigma^j(b))_j$$

is an isomorphism: after reduction it is the usual isomorphism for F'/F_0 , and both sides are free R'_b -modules of rank n .

On $\mathcal{A}_{R'_b}$ define the semilinear descent action $\delta(x) = \sigma(x) + \tilde{P}$. Here σ acts on the coefficients, whereas δ also translates the elliptic-curve point. Its n th power is the identity. The divisor

$$D' = \sum_{j=0}^{n-1} [j\tilde{P}]$$

is invariant. Its sections are disjoint because their reductions are distinct. The line bundle $\mathcal{O}(3D')$ is a tensor product of translates of the relatively ample Weierstrass line bundle $\mathcal{O}(3[0])$. It is relatively ample, and therefore so is $\mathcal{O}(D')$. The invariant divisor supplies its canonical compatible linearization. Finite *étale* Galois descent (Foundation 6) gives $\mathcal{X}$ and D , with a splitting identification $\mathcal{X}_{R'_b} \simeq \mathcal{A}_{R'_b}$. Their reduction is precisely the twist E and the corresponding origin-orbit divisor. Smoothness and relative ampleness descend.

Under this identification, continue to write ξ_i for the corresponding sections of $\mathcal{X}_{R'_b}$. They have distinct reductions, all different from the constant points jP . They are therefore pairwise disjoint and avoid D' . Under the descent action they satisfy $\delta(\xi_i) = \xi_{i+1}$. Their union is an invariant disjoint union of sections, hence a finite *étale* divisor of degree n , and descends with these properties, although the individual sections need not be defined over R_b . $\square$

6.2 A spherical base and its finite extensions

For a valued field we write the valuation additively, with $v(0) = +\infty$. A *closed ball* of radius δ means $B(a, \delta) = \{x : v(x - a) \geq \delta\}$, where δ belongs to the value group. A valued field is *spherically complete* if every nested family of nonempty closed balls has nonempty intersection. An extension is *immediate* if its value group and residue field are unchanged.

The purpose of the next construction is twofold. Finite field degrees will equal full residue degrees, so a small-orbit degree bound survives reduction. Finite-dimensional section spaces will also have free integral lattices with full-dimensional reduction, allowing comparison of the generic and residue genera. The construction uses the classical maximal-immediate-extension and pseudo-limit methods of Krull and Kaplansky [26, 23]; see [32, Section 5] for an accompanying account. We give the details for our specified residue field, without assuming it perfect or algebraically closed.

Proposition 6.2. *There is an extension K_s/K_b with value group $\mathbb{Q}$ and residue field F_0 which is spherically complete. Every finite extension L/K_s , equipped with an extension of its valuation, has value group $\mathbb{Q}$, is spherically complete, and satisfies*

$$[\text{res}(L) : F_0] = [L : K_s]. \quad (6.1)$$

The valuation on K_s , and on each such L , has a unique extension to every algebraic extension. Moreover $\overline{K_s} \simeq \mathbb{C}$ as abstract fields.

We prove the construction and the linear-algebra assertions first, and establish uniqueness after proving simultaneous residue approximation.

Construction of K_s . Normalize the valuation of Proposition 6.1 by $v(\varpi) = 1$. In an algebraic closure with an extended valuation, choose compatible factorial roots: take $\theta_1 = \varpi$ and $\theta_{j+1}^{j+1} = \theta_j$ for $j \geq 1$. Thus $\theta_j^{j!} = \varpi$ and the fields $K_b(\theta_j)$ are nested. Put $K_\infty = \bigcup_{j \geq 1} K_b(\theta_j)$. For $r = j!$ and $\theta = \theta_j$, one has $\theta^r = \varpi$. The values of $1, \theta, \dots, \theta^{r-1}$ lie in distinct cosets of $\mathbb{Z}$. These elements are linearly independent over K_b , and the equation for θ shows that they are a basis of $K_b(\theta)$. For $x = \sum_{j=0}^{r-1} a_j \theta^j \neq 0$, the distinct value cosets give

$$v(x) = \min_j (v(a_j) + j/r).$$

The value group is $r^{-1}\mathbb{Z}$. If $v(x) = 0$, the minimum can only occur at $j = 0$, and all other terms have positive value; hence the residue of x is the residue of a_0 . This proves that K_∞ has value group $\mathbb{Q}$ and residue field F_0 .

Here is the set-theoretic detail needed to choose a maximal immediate extension. Any valued field with countable value group and countable residue field has cardinality at most $\mathfrak{c} = 2^{\aleph_0}$. Indeed, for every q in the value group and every closed q -ball B , the open q -balls contained in B are in bijection with the residue field, after choosing a center and an element of value q . Label these open subballs injectively by natural numbers. For a point x , record, for each q , the label of its open q -ball inside its closed q -ball. If $x \neq y$, at $q = v(x - y)$ they belong to the same closed ball but different open subballs, so these records differ. This gives an injection into a countable product of countable sets. In our situation F_0 is countable, being a subfield of the finitely generated field F'/κ .

Choose a fixed set of cardinality greater than $\mathfrak{c}$, containing the underlying set of K_∞ . Order the immediate valued extensions of K_∞ carried by subsets of this set by literal extension of all their structures. The union of a chain is an immediate valued field: each of its elements and every finite field computation occur in one member of the chain. The cardinality bound applies to the union as

well. Zorn's lemma gives a maximal member K_s . Any further immediate extension has cardinality at most $\mathfrak{c}$ and could be relabeled, fixing K_s , into the unused part of the fixed set. Thus K_s has no proper immediate extension in the unrestricted sense.

We now prove spherical completeness directly. Suppose a nest of nonempty closed balls has empty intersection. Distinct balls in a nest with the same radius coincide, so the countability of the value group gives a countable cofinal subnest. Successively go deeper in that subnest, also choosing each next ball to omit the preceding center. We obtain cofinal balls $B(a_i, \delta_i)$ such that $a_i \notin B(a_{i+1}, \delta_{i+1})$ and

$$v(a_j - a_i) = \gamma_i \quad (j > i), \quad \delta_i \leq \gamma_i < \delta_{i+1}, \quad \gamma_1 < \gamma_2 < \cdots. \quad (6.2)$$

The balls $B(a_i, \gamma_i)$ are nested, lie inside $B(a_i, \delta_i)$, and still have empty intersection.

Extend the valuation to an algebraic closure of K_s . Call b a *pseudo-limit* of this sequence if $v(b - a_i) = \gamma_i$ for all sufficiently large i . No element of K_s is a pseudo-limit, since a pseudo-limit belongs to every ball $B(a_i, \gamma_i)$. For any point b which is not a pseudo-limit, the values $v(b - a_i)$ are eventually constant and smaller than γ_i . To see this, if $v(b - a_i) < \gamma_i$ at one stage, the strong triangle law and (6.2) give the same value at every later stage. If $v(b - a_i) > \gamma_i$, the next value is γ_i , which is smaller than γ_{i+1} , reducing to the preceding case. Otherwise equality holds eventually, as required for a pseudo-limit.

Suppose first that an algebraic pseudo-limit α exists, and choose one of smallest degree d over K_s . Every root β of a nonzero polynomial $f \in K_s[T]$ with $\deg f < d$ is not a pseudo-limit. For large i we therefore have

$$v(a_i - \beta) = v(\alpha - \beta) < v(\alpha - a_i), \quad \frac{\alpha - \beta}{a_i - \beta} \equiv 1 \pmod{\mathfrak{m}_{\overline{K_s}}}.$$

Factoring f over the algebraic closure yields

$$f(\alpha)/f(a_i) \equiv 1 \pmod{\mathfrak{m}_{\overline{K_s}}} \quad (i \gg 0). \quad (6.3)$$

Every nonzero element of $K_s(\alpha)$ is $f(\alpha)$ for a polynomial of degree less than d . Equation (6.3) therefore gives, for each such element x , a scalar $c = f(a_i) \in K_s$ with $v(x - c) > v(x)$. It follows that $v(c) = v(x)$, so no new values occur. If $v(x) = 0$, the same inequality says that x and c have the same residue, so no new residue-field elements occur either. Thus $K_s(\alpha)/K_s$ is a proper immediate extension, a contradiction.

Suppose instead that no algebraic pseudo-limit exists. Factoring any nonzero polynomial over the fixed algebraic closure shows that its values at a_i are eventually constant and finite; the same holds for nonzero rational functions. Define, on $K_s(T)$,

$$w(f) = \text{the eventual value of } v(f(a_i)), \quad w(0) = +\infty.$$

Evaluation proves multiplicativity and the strong triangle inequality, so w is a valuation extending that of K_s . For fixed $f \neq 0$ and $j > i \gg 0$, every linear factor in its numerator and denominator satisfies $(a_j - \beta)/(a_i - \beta) \equiv 1 \pmod{\mathfrak{m}_{\overline{K_s}}}$, where β is its root. Indeed $v(a_i - \beta) < \gamma_i = v(a_j - a_i)$. Consequently

$$v(f(a_j) - f(a_i)) > v(f(a_i)) = w(f).$$

For a fixed sufficiently large i , the left side has an eventual value as j increases, since it evaluates the rational function $f(T) - f(a_i)$. Thus $w(f - f(a_i)) > w(f)$ unless the difference is zero, when the assertion is immediate. Thus every nonzero element f of $K_s(T)$ admits a scalar $f(a_i)$ with this strict approximation inequality. As in the algebraic case, this shows that neither the value group nor the residue field changes. This is again a proper immediate extension, contradicting maximality. Both cases being impossible, K_s is spherically complete. $\square$

Lemma 6.3. *Let K_* be a spherically complete field with value group $\mathbb{Q}$, valuation ring R_* , maximal ideal $\mathfrak{m}_*$, and residue field k_* . Let V be a finite-dimensional K_* -vector space with a valuation norm $\lambda : V \rightarrow \mathbb{Q} \cup \{+\infty\}$: it is finite away from zero, has $\lambda(0) = +\infty$, satisfies the strong triangle inequality, and obeys $\lambda(ax) = v(a) + \lambda(x)$ for $a \neq 0$. There is a basis $e_1, \dots, e_d$ of norm zero such that*

$$\lambda\left(\sum_i a_i e_i\right) = \min_i v(a_i). \quad (6.4)$$

In particular its integral and positive balls are

$$V_{\geq 0} = \bigoplus_i R_* e_i, \quad V_{> 0} = \bigoplus_i \mathfrak{m}_* e_i = \mathfrak{m}_* V_{\geq 0}, \quad \dim_{k_*}(V_{\geq 0}/V_{> 0}) = d. \quad (6.5)$$

The normed space V is spherically complete.

Proof. A subspace with an orthogonal basis is spherically complete: its closed balls are products of closed balls in the finitely many coordinates, with the radii shifted by the norms of the basis vectors. Each coordinate nest has a common point.

Suppose an orthogonal basis has already been constructed for a subspace U , and choose $x \notin U$. For every value γ achieved by $\lambda(x - u)$, $u \in U$, consider

$$B_\gamma = \{u \in U : \lambda(x - u) \geq \gamma\}.$$

This is a nonempty closed ball in U : if u_γ belongs to it, then it equals $\{u : \lambda(u - u_\gamma) \geq \gamma\}$. These balls form a nest, so they have a common point u_0 . The finite value $\ell = \lambda(x - u_0)$ is at least every achieved value and itself is achieved. Thus u_0 is a best approximation to x . Put $y = x - u_0$. For $u \in U$, maximality of ℓ gives $\lambda(y + u) \leq \ell$. If $\lambda(u) < \ell$, the strong triangle law gives $\lambda(y + u) = \lambda(u)$; if $\lambda(u) \geq \ell$, it gives $\lambda(y + u) \geq \ell$, hence equality. Therefore

$$\lambda(y + u) = \min\{\ell, \lambda(u)\}.$$

Homogeneity shows that adjoining y preserves orthogonality. Induction gives an orthogonal basis of V . Since every basis norm is in the base value group, scale its vectors to norm zero. Formulae (6.4) and (6.5) follow, and the same coordinate argument proves spherical completeness of V . $\square$

Finite-extension assertions of Proposition 6.2. For any algebraic extension of a field with value group $\mathbb{Q}$, every nonzero algebraic element has value in $\mathbb{Q}$. Indeed, in a polynomial relation $\sum_i a_i x^i = 0$, at least two nonzero terms must have the same minimum value, so

$$(i - j)v(x) = v(a_j) - v(a_i) \in \mathbb{Q} \quad (i \neq j).$$

Divisibility of $\mathbb{Q}$ and torsion-freeness of ordered value groups imply $v(x) \in \mathbb{Q}$. Apply Lemma 6.3 to a finite extension L/K_s with any extended valuation, viewed as a normed vector space over K_s . Its integral and positive balls are exactly its valuation ring and maximal ideal. Formula (6.5) proves (6.1), including the full inseparable degree of the residue extension. Orthogonal coordinates also show that L is spherically complete. The same argument applies to every finite extension of L . $\square$

The preceding argument gives equality of field and residue degrees for finite extensions of the spherical base. We also need a statement for valued fields that are not assumed spherically complete, where several prolongations can occur. The next lemma supplies the residue-degree bound used for curve function fields in Section 7 and completes the uniqueness assertion for the spherical base.

Lemma 6.4 (Simultaneous residue approximation). *Let T be a nontrivially valued field with value group $\mathbb{Q}$ and infinite residue field k_T . Let U/T be finite and let $w_1, \dots, w_r$ be distinct prolongations of its valuation, with residue fields $k_1, \dots, k_r$. Then*

$$\bigcap_{i=1}^r \mathcal{O}_{w_i} \longrightarrow \prod_{i=1}^r k_i$$

is surjective, and

$$\sum_{i=1}^r [k_i : k_T] \leq [U : T]. \quad (6.6)$$

Proof. The preceding polynomial-relation argument puts all the values in $\mathbb{Q}$, with their normalization fixed by the valuation on T . For $i \neq j$, distinctness gives an element whose two values differ. Inverting it if necessary and multiplying by an element of T of a suitable rational value produces $z \in U$ with $w_i(z) > 0 > w_j(z)$. Choose a unit $c \in T$ such that, at every w_ℓ for which $w_\ell(z) = 0$, the residue of $1 + cz$ is nonzero. Each such condition excludes at most one value for the residue of c in $k_T^\times$, so the infinitude of k_T permits this choice. Then

$$q_{ij} = \frac{1}{1 + cz}$$

is integral at every w_ℓ , has residue 1 at i , and has residue 0 at j . Indeed, a positive value of z makes the denominator a unit with residue 1; a negative value of z gives the denominator that negative value; and value zero is covered by the choice of c .

Put $q_i = \prod_{j \neq i} q_{ij}$, with $q_i = 1$ if $r = 1$. It has residue 1 at i and positive value at all other places. For a prescribed residue $b_i \in k_i$, choose any lift $\tilde{b}_i$ integral at i . If the lift is nonzero, choose N_i so that

$$N_i w_\ell(q_i) + w_\ell(\tilde{b}_i) > 0 \quad (\ell \neq i).$$

There are only finitely many conditions, and every $w_\ell(q_i)$ in them is positive. At i , the factor $q_i^{N_i}$ has residue one. Thus $q_i^{N_i} \tilde{b}_i$ is integral everywhere, has residue b_i at i , and has residue zero at every other place. A zero lift contributes the zero term. Summing these terms realizes any tuple $(b_i)_i$.

For completeness, each $[k_i : k_T]$ is finite and at most $[U : T]$ by the single-place lifting argument already used in the proof of [Proposition 6.1](#). Choose a k_T -basis of each k_i , and use the surjectivity just proved to lift its elements with zero residues in the other coordinates. These lifts are linearly independent over T . For if they satisfied a nonzero relation, divide it by a coefficient of minimum value. Every coefficient then is integral and at least one is a unit. Reducing in a coordinate which contains such a unit contradicts the chosen residue basis there. Their total number is therefore at most $[U : T]$, proving (6.6). $\square$

Completion of the proof of Proposition 6.2. The residue of K_s , and that of every finite extension L/K_s , is infinite. If two distinct valuations extended the valuation of L to a finite extension U/L , the finite-extension argument above would give residue degree $[U : L]$ for each. This contradicts (6.6). Existence of prolongations is part of [Foundation 5](#); uniqueness on finite extensions implies uniqueness on arbitrary algebraic extensions by restriction to finite subextensions.

Finally $|K_s| \leq \mathfrak{c}$ by the cardinality argument, while $\mathbb{Q}_p \subset K_s$ gives $|K_s| \geq \mathfrak{c}$. Its algebraic closure has the same cardinality. An uncountable algebraically closed field of characteristic zero has cardinality equal to its transcendence degree over $\mathbb{Q}$: the field generated by a transcendence basis, and then its algebraic closure, both have cardinality the maximum of that basis cardinality and $\aleph_0$. Both $\overline{K_s}$ and $\mathbb{C}$ therefore have transcendence degree $\mathfrak{c}$ over $\mathbb{Q}$. The algebraically closed field classification in [Foundation 4](#) gives the asserted abstract isomorphism. No compatibility with either field's topology is asserted or used. $\square$

6.3 Marked covers and a small orbit

Choose an embedding $K'_b \hookrightarrow \overline{K_s}$ over K_b and an abstract isomorphism $\overline{K_s} \simeq \mathbb{C}$. Write $X_s = \mathcal{X}_{K_s}$. The embedding carries the sections ξ_i on the split model $\mathcal{X}_{R'_b}$ to the specified geometric marked points on X_s . A *marked J -cover* of $\mathcal{X}_{\overline{K_s}}$ will mean a connected smooth projective curve Y with a finite map to $\mathcal{X}_{\overline{K_s}}$ which is Galois of group J , together with a specified left action of J by deck transformations. Isomorphisms are over the fixed base curve and commute with every element of this specified action.

Lemma 6.5. *After choosing oriented handle and puncture generators on the complex punctured curve, isomorphism classes of marked J -covers branched only at $\xi_1, \dots, \xi_n$ correspond bijectively to*

$$\left\{ (x, y, u_1, \dots, u_n) \in J^{n+2} : \begin{array}{l} [x, y]u_1 \cdots u_n = 1, \\ \langle x, y, u_1, \dots, u_n \rangle = J \end{array} \right\} / J,$$

where J acts by simultaneous inner conjugation. The inertia order at ξ_i is $|u_i|$. For a prescribed multiset $\mathcal{C}$ of n J -conjugacy classes of p -singular elements, let $\mathcal{T}_{\mathcal{C}}$ be the finite set of these marked-cover classes having that multiset of local classes. Then $\mathcal{T}_{\mathcal{C}}$ is preserved by $\text{Gal}(\overline{K_s}/K_s)$.

Thus $|\mathcal{T}_{\mathcal{C}}| = N_{J,s}(\mathcal{C})$ in the notation of [Theorem 3.1](#). The labels $\xi_1, \dots, \xi_n$ remain ordered: prescribing their class multiset does not divide out by permutations of the labels.

Proof. Under the chosen complex identification the base is a smooth Weierstrass cubic. In characteristic zero, completing the square gives an equation $y^2 = f_3(x)$ with three distinct roots. Its double cover of the projective line branches simply at those roots and at infinity. Cut the sphere along two disjoint arcs joining pairs of these four points and glue the two sheets crosswise. The resulting compact surface is a torus. This identifies the topological type of our particular base without a general analytic–algebraic genus comparison.

Remove the n marked points. Its complex fundamental group has generators and relation $[x, y]u_1 \cdots u_n = 1$ as in [Foundation 12](#). Choose a point in a fiber of a marked cover and identify the fiber with the regular left J -set. Path lifting commutes with the left deck action, so monodromy acts by right translations. With path multiplication chosen in the order of successive traversal, these translations specify a homomorphism from the fundamental group to J . The cover is connected exactly when the monodromy action is transitive, which for this regular action means that the homomorphism is surjective. Changing the chosen fiber point conjugates this homomorphism by one element of J . Conversely, a surjective homomorphism defines such a fiber action and therefore a connected unramified cover. An equivariant fiber bijection is a right translation, and compatibility with monodromy says exactly that the two homomorphisms are simultaneously conjugate. Thus the quotient is by inner conjugation; keeping the specified J -action does not identify covers through outer automorphisms of J .

Riemann existence in [Foundation 12](#) algebraizes the unramified cover of this punctured algebraic curve, with its morphisms and specified deck action. Normalize the original projective base in the resulting function field. Finiteness of normalization gives a projective algebraic cover; normality makes its source smooth in characteristic zero. The deck action and isomorphisms extend uniquely across normalization. In a local power-map chart $z \mapsto z^e$, the ramification index is e , the order of the associated monodromy element. These extensions give precisely the branched covers and isomorphisms in the statement.

We describe the local conjugacy class algebraically in order to check Galois invariance. For each divisor e of m , use the root $\zeta_e \in K_s$ which maps to $\exp(2\pi i/e)$ under the chosen complex identification. All these roots belong to K_s by [Proposition 6.1](#). Represent the based puncture loop

as an approach path, a positive circle around the puncture, and the reverse approach path. In the chart $z \mapsto z^e$, lifting the positive circle sends z to $\zeta_e z$. Let Q be the point above the puncture selected by the lifted approach path, and let $g \in J$ be the inertia element acting this way in its local chart. Apply g to the lifted approach path. The reverse of the resulting path is the lift of the return path starting at the endpoint of the circle lift. Thus the complete based loop sends the chosen fiber point to its image under g . Under the preceding fiber identification, its monodromy element is therefore g . The inertia group is cyclic, and g is its unique generator acting on the tangent line at Q by ζ_e . Choosing another point above the puncture conjugates the element in J . This recovers its conjugacy class from the algebraic tangent action. (Reversing all local-loop conventions replaces all these generators by their inverses consistently.)

A Galois automorphism over K_s sends the tangent action at Q to the tangent action at its conjugate point, and fixes ζ_e . Consequently it preserves the local J -conjugacy class at the transported branch point. It permutes the branch points, since their union is defined over K_s , and hence preserves their prescribed class multiset and the condition that all inertia orders are divisible by p . This proves invariance of $\mathcal{T}_C$. Its finiteness follows from the finite tuple description. Although Galois can permute the marked points, an isomorphism of the marked covers is over the fixed geometric base curve. The cover classes are therefore still parametrized by simultaneous inner conjugacy, without quotienting by those permutations. $\square$

For a cover with its group action specified, the obstruction to descent is center-valued; compare the marked-cover analysis of Dèbes and Douai [10, Section 3.2]. In the following proposition we realize that obstruction as an explicit finite group extension and split it by a Sylow argument. Thus a field of moduli is not silently assumed to be a field of definition.

Proposition 6.6 (Descent from a small Sylow orbit). *Assume $O_p(J) = 1$ and fix a multiset $\mathcal{C}$ as in Lemma 6.5. There is a finite Galois extension M/K_s through which the action on $\mathcal{T}_C$ factors. Let P_Γ be a Sylow p -subgroup of $\Gamma = \text{Gal}(M/K_s)$. If a P_Γ -orbit has size less than p^h , one of its marked covers descends to a finite extension L/K_s satisfying*

$$[L : K_s]_p < p^h, \quad [F : F_0]_p = [L : K_s]_p, \quad F = \text{res}(L). \quad (6.7)$$

The descended map $Y \rightarrow X = \mathcal{X}_L$ has geometrically connected smooth projective source, degree m , the specified J -action, and all branch indices at the ξ_i divisible by p .

Proof. Choose a representative for each of the finitely many marked-cover classes. Their projective equations, maps, and specified deck actions are finite data, and hence are defined over one finite extension of K_s by Foundation 6. Take a finite Galois extension N/K_s containing these fields of definition, and enlarge it if necessary to contain $K_s K'_b$. For every $\tau \in \text{Gal}(N/K_s)$ and every chosen representative, choose a marked isomorphism between its τ -conjugate and the representative of the resulting class. Such an isomorphism exists over $\overline{K_s}$ by Lemma 6.5; there are only finitely many choices to make. Their finite coefficient data lie in a further finite extension, which we enclose in a finite Galois extension M/K_s . This two-stage choice ensures that all required isomorphisms are defined over M . Conjugates of the original models under $\text{Gal}(M/K_s)$ depend only on restriction to N and are scalar extensions of the already considered conjugates. The action on classes therefore factors through Γ .

Let $S_p \leq P_\Gamma$ stabilize a class in a small orbit and let Y_M be its chosen model. Since S_p fixes this isomorphism class, for each $\tau \in S_p$ the chosen isomorphisms supply a τ -semilinear isomorphism of Y_M over the corresponding action on $\mathcal{X}_M$, commuting with the specified J -action. Here semilinear means that the map changes scalars by τ rather than fixing M . These maps have not been chosen to respect multiplication in S_p ; that compatibility is the remaining descent problem.

Form the group $\mathcal{E}$ of all such marked semilinear isomorphisms, for all $\tau \in S_p$, with composition as its group operation. Projection to the scalar action is a surjection $\mathcal{E} \rightarrow S_p$. An automorphism of Y_M over $\mathcal{X}_M$ is a deck transformation: after extension to $\overline{K_s}$ the Galois cover has exactly its m specified deck transformations, all already defined over M . Those commuting with the specified J -action are exactly $Z(J)$. Every marked semilinear map also commutes with these specified deck transformations. Thus this kernel is central, and $\mathcal{E}$ is finite and fits into an exact sequence

$$1 \longrightarrow Z(J) \longrightarrow \mathcal{E} \longrightarrow S_p \longrightarrow 1. \quad (6.8)$$

The Sylow p -subgroup of the abelian group $Z(J)$ is characteristic there, hence normal in J . Since $O_p(J) = 1$, it is trivial; thus $p \nmid |Z(J)|$. A Sylow p -subgroup of $\mathcal{E}$ has order $|S_p|$, intersects $Z(J)$ trivially, and therefore maps isomorphically onto S_p . Its inverse supplies a section of (6.8), assigning maps ϕ_τ with

$$\phi_\tau \phi_\rho = \phi_{\tau\rho}, \quad \phi_1 = \text{id}.$$

These are exactly the compatibility identities for a descent datum on Y_M and its cover map. Each ϕ_τ commutes with every specified element of J , so the datum descends the marked action as well.

Take the pullback of an ample line bundle on X_s , for example $\mathcal{O}(D)$ on its generic fiber. It is ample on Y_M , and the semilinear cover maps give its compatible linearization. Galois descent in [Foundation 6](#) therefore gives the marked cover over $L = M^{S_p}$. Its claimed geometric properties can be checked after extension to $\overline{K_s}$, where they are those of the chosen cover. In particular its function-field extension is Galois of degree m with the specified group J . Finally

$$[L : K_s]_p = [\Gamma : S_p]_p = \frac{|P_\Gamma|}{|S_p|} = |P_\Gamma : S_p| < p^h.$$

Equation (6.1) gives the residue assertion in (6.7). □

6.4 Matching branch permutations after reduction

In [Section 7](#) we will impose vanishing over selected marked points. The same subset of label indices must define a branch subset over the characteristic-zero field and over its residue field. We therefore compare their Galois permutation actions. These are the actions on the marked curve; in split elliptic-curve coordinates they include the translating torsion point in the descent data.

Lemma 6.7. *Let L/K_s be finite, with residue F , and set $L' = LK'_b$. In the residue field of L' , form $S = FF'$ using the residues of its two subfields. Then L'/L and S/F are finite Galois extensions, the residue field of L' is exactly S , and reduction induces an isomorphism*

$$\text{Gal}(L'/L) \simeq \text{Gal}(S/F).$$

Under this isomorphism the permutation actions on $\xi_1, \dots, \xi_n$ and on $t_1, \dots, t_n$ agree. In particular, every subset of indices invariant under these identified permutation groups defines a branch subset over L and over F , respectively.

Proof. The uniqueness proved in [Proposition 6.2](#) gives one valuation on L' . Its restriction to K'_b is the valuation of [Proposition 6.1](#), by uniqueness over the complete discretely valued field K_b . Put $T = \text{res}(L')$. It contains both F and F' , so it contains their compositum S . Base change of the finite Galois extension K'_b/K_b gives the Galois extension L'/L , and restriction embeds $H = \text{Gal}(L'/L)$ in $\text{Gal}(K'_b/K_b)$. Every element of H preserves the unique valuation and hence acts on T , fixing F and preserving F' . Its induced action on S is faithful: if it is trivial on S , it is trivial on F' ; the faithful

residue action of $\text{Gal}(K'_b/K_b)$ then makes its restriction to K'_b trivial; and L and K'_b generate L' . Thus reduction injects H into $\text{Aut}_F(S)$. Also S/F is Galois, being a compositum with the finite Galois extension F'/F_0 . The residue-degree bound now gives the complete degree comparison

$$|H| \leq [S : F] \leq [T : F] \leq [L' : L] = |H|. \quad (6.9)$$

All these inequalities are equalities. In particular $T = S$ and the injection of groups is an isomorphism.

The sections ξ_i extend over the valuation ring of L' by base change from the split good-reduction model, and reduce to t_i . Their reductions are distinct. Applying a member of H to a section and then reducing gives the same section as first reducing and then applying its image in $\text{Gal}(S/F)$. This proves equality of the two permutations of the index set. An invariant subset of a split finite *étale* branch divisor descends along the respective finite Galois extension. The final assertion follows by taking the same invariant subset of indices under these identified permutation actions: it selects the ξ_i on the generic side and the t_i on the residue side. $\square$

Figure 2 records these inclusions and residue fields in the situation of Proposition 6.6.

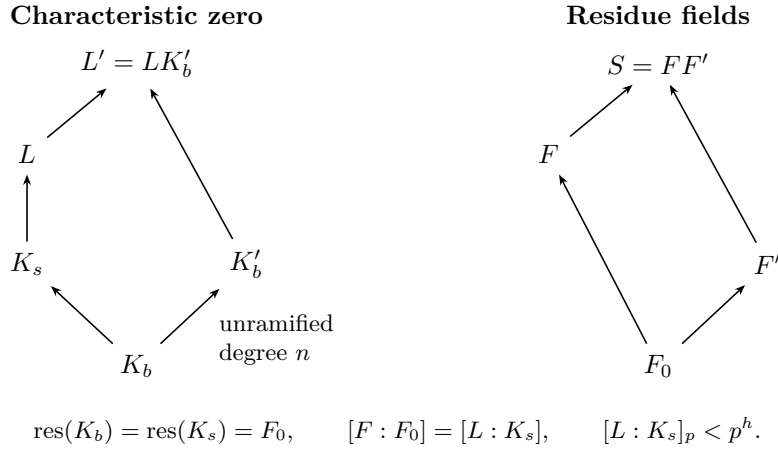


Figure 2: The auxiliary fields in a descent arising from a small Sylow orbit. Arrows denote field inclusions. The right-hand diagram records the residues of the left-hand fields; in particular K_b and K_s have the same residue. The field K'_b splits the marked sections, and its residue F' splits their reductions. The cyclic extension K'_b/K_b reduces to F'/F_0 , and the Galois actions of L'/L and S/F induce the same permutations of the marked points.

7 Specialization, connectedness, and the tuple count

We retain the auxiliary fields and marked genus-one curve constructed in Propositions 6.1 and 6.2. In this section $m = |J|$, $h \geq 1$, and $n = p^s$. Choose s so that

$$s \geq \max\{1, h\}, \quad p^{s-h} > m(h + \log_p m). \quad (7.1)$$

This is a sufficient threshold for Theorem 5.1. The relatively ample effective divisor D has degree n on each fiber and is disjoint from the marked sections after their splitting extension.

Proposition 7.1. *Suppose $O_p(J) = 1$ and (7.1) holds. There is no finite extension L/K_s with $[L : K_s]_p < p^h$ over which the marked curve $X = \mathcal{X}_L$ admits a geometrically connected smooth*

projective J -Galois cover $f : Y \rightarrow X$, with its J -action defined over L , branched only at the marked points and with ramification index divisible by p at every marked point.

We will obtain the contradiction by producing a nontrivial normal p -subgroup of J . To this end, we first extract residue curves from one J -orbit of valuations of a supposed cover. A comparison of sections determines their total degree. A second comparison, with prescribed vanishing, combines with the genus estimates to force equality of Euler characteristics. These two equalities yield a flat model whose special fiber is the disjoint union of those residue curves. Connectedness will force the orbit to have one member; the kernel of the resulting J -action on its residue field will be the required nontrivial normal p -subgroup.

Suppose, for the proof, that such a cover exists. Its smooth, geometrically connected source Y is geometrically integral: over an algebraic closure, the irreducible components of a smooth curve are disjoint, so connectedness leaves only one. Let R be the valuation ring of L , with maximal ideal $\mathfrak{m}_R$ and residue field F . By [Proposition 6.2](#), L is spherically complete, its value group is $\mathbb{Q}$, and

$$[F : F_0]_p = [L : K_s]_p < p^h.$$

Put $L' = LK'_b$ and $S = FF'$. By [Lemma 6.7](#), S is the residue field of L' , and the permutation actions on the labels ξ_i over L' and t_i over S agree. Write D_X and D_F for the divisors induced by D on X and E_F , respectively, and put $D_Y = f^*D_X$. The field extension $L(Y)/L(X)$ is Galois of degree m , with group J : these assertions hold after extension to an algebraic closure, and the specified automorphisms are already defined over L .

Let e_i be the geometric ramification index at ξ_i . Each e_i is divisible by p . Riemann–Hurwitz over an algebraic closure, together with invariance of Euler characteristic under field extension, gives

$$\frac{\nu(Y)}{m} = \frac{-2\chi(Y)}{m} = \sum_{i=1}^n \left(1 - \frac{1}{e_i}\right). \quad (7.2)$$

Here the base has genus one, and the reduced geometric fiber above ξ_i consists of m/e_i points. The characteristic-zero form of Riemann–Hurwitz being used is included in [Foundation 12](#).

7.1 The Gauss valuation and reduction of sections

Lemma 7.2. *There is a valuation v_X on $L(X)$ extending that on L , with value group $\mathbb{Q}$ and residue field $F(E_F)$, having the following properties. For every sufficiently large j , the integral ball in $H^0(X, \mathcal{O}_X(jD_X))$ reduces onto $H^0(E_F, \mathcal{O}_{E_F}(jD_F))$. Reduction is compatible with inclusions as j increases and with multiplication. Moreover, for an integral section b in such a space,*

$$\overline{b(\xi_i)} = \bar{b}(t_i) \quad \text{in } S. \quad (7.3)$$

In particular, vanishing at ξ_i implies vanishing of its reduction at t_i .

Proof. Work initially over the complete DVR R_b , and write $D_E = D|_E$. For j sufficiently large, [Foundation 9](#) and the special-fiber sequence give

$$H^0(\mathcal{X}, \mathcal{O}(jD)) \otimes_{R_b} F_0 \simeq H^0(E, \mathcal{O}_E(jD_E)).$$

Indeed Serre vanishing kills $H^1(\mathcal{X}, \mathcal{O}(jD))$, so the sequence obtained by multiplication by a uniformizer gives surjectivity onto the special sections. The module on the left before reduction is finite and torsion-free, hence free over R_b ; torsion-freeness follows from flatness of $\mathcal{X}$ and invertibility of

$\mathcal{O}(jD)$. Choose a basis $b_{j,1}, \dots, b_{j,t_j}$ whose reductions are a basis of the special section space. Flat base change supplies bases over L on the generic side and over F on the special side.

As D_X is effective, regard these sections as rational functions. For $b = \sum_{\alpha} c_{\alpha} b_{j,\alpha}$ put

$$v_X(b) = \min_{\alpha} v(c_{\alpha}).$$

Integral coefficients give a special rational function with poles bounded by jD_F , and this function is nonzero if the displayed minimum is zero. The inclusions for $jD \leq j'D$ are integral maps on the original model and reduce to injective inclusions of special section spaces. Consequently the assigned value is independent of the sufficiently large space containing b : scale the coefficients so that their minimum is zero, and use the nonzero special image. The same argument proves multiplicativity. After scaling two nonzero sections to value zero, their reductions are nonzero, and their product is nonzero because E_F is integral. The strong triangle inequality follows directly from the coefficient rule.

The union of these section spaces is a ring whose fraction field is $L(X)$: sufficiently high ample multiples give a projective embedding, and their section ratios generate the function field. Thus the rule extends to a valuation on $L(X)$. Its values lie in $\mathbb{Q}$, and all values in $\mathbb{Q}$ occur already on L . A fraction of value zero can be written with numerator and denominator both of value zero, by a common scalar rescaling. Its residue is the ratio of their nonzero special reductions, hence lies in $F(E_F)$. Conversely the ratios of high-degree special sections generate $F(E_F)$, and each such section has an integral lift. This identifies the residue field.

Finally the basis functions are regular along the splitting sections over R'_b , since those sections avoid D . Their values are integral and reduce to their evaluations at t_i . The same holds after extension of coefficients from R_b to R and evaluation over the valuation ring of L' . Since its residue is S , this proves (7.3). $\square$

Choose one prolongation of v_X to $L(Y)$, and let $\mathcal{V}$ be its orbit of distinct conjugates under J . For $w \in \mathcal{V}$ let $K(w)$ be its residue field. The single-place residue-degree bound in Lemma 6.4 makes $K(w)/F(E_F)$ finite; the Gauss valuation has value group $\mathbb{Q}$ and infinite residue field $F(E_F)$. Set

$$m_w = [K(w) : F(E_F)].$$

Let C_w be the normalization of E_F in $K(w)$ and D_w the pullback of D_F . These are regular integral projective curves, with finite maps to E_F , by Foundation 7. Throughout the next argument we use this single orbit; its size is not assumed in advance to account for all the extension degree.

For $j \geq 0$ put

$$V_j = H^0(Y, \mathcal{O}_Y(jD_Y)), \quad v_{\mathcal{V}}(a) = \min_{w \in \mathcal{V}} w(a), \quad A_j = \{a \in V_j : v_{\mathcal{V}}(a) \geq 0\}.$$

The minimum is a valuation norm on the L -vector space V_j . All prolongation values lie in $\mathbb{Q}$: algebraicity makes the extension of value groups torsion, while the base value group is divisible. The orthogonal-basis assertion of Lemma 6.3 therefore applies, and an orthogonal basis can be scaled to have value zero. In this basis,

$$A_j \simeq R^{\dim_L V_j}, \quad \{a \in V_j : v_{\mathcal{V}}(a) > 0\} = \mathfrak{m}_R A_j. \quad (7.4)$$

These assertions also hold on any L -linear subspace of V_j . At this point reduction means the map of rational functions

$$A_j \longrightarrow \prod_{w \in \mathcal{V}} K(w), \quad a \longmapsto (\bar{a}_w)_w.$$

Its kernel is the positive ball $\mathfrak{m}_R A_j$, so its image has dimension $\dim_L V_j$ over F . No model of Y has been used. The next lemma locates these residue-field elements in the section spaces of the separately normalized curves C_w .

Lemma 7.3. *For all sufficiently large j , simultaneous residue gives an injection*

$$A_j/\mathfrak{m}_R A_j \hookrightarrow \bigoplus_{w \in \mathcal{V}} H^0(C_w, \mathcal{O}_{C_w}(jD_w)). \quad (7.5)$$

These maps respect products. If a section vanishes on the reduced inverse images of a selected set of marked points, its reductions vanish on the corresponding reduced inverse images on the C_w , provided the selections are defined over the respective ground fields. The vanishing assertion may be checked after splitting the marked points over L' and S .

Proof. Let $a \in A_j$. Form the polynomial using every element of J , including any repeated conjugates:

$$P_a(T) = \prod_{g \in J} (T - g(a)) = T^m + c_1 T^{m-1} + \cdots + c_m. \quad (7.6)$$

The coefficients are J -invariant rational functions, hence belong to $L(Y)^J = L(X)$. At a fixed $w \in \mathcal{V}$ every $g(a)$ is integral, because g permutes $\mathcal{V}$. Thus each c_k has nonnegative Gauss value. Moreover

$$c_k \in H^0(X, \mathcal{O}_X(kjD_X)).$$

To see the pole bound, each conjugate has poles bounded by jD_Y , so the elementary symmetric function of degree k has poles bounded by kjD_Y . Since the coefficient is a rational function on X , its pole bound on X follows from that on its finite surjective pullback.

By Lemma 7.2, $\bar{c}_k$ has poles bounded by kjD_F . The residue $\bar{a}_w \in K(w)$ satisfies the reduction of (7.6). Let t be a local equation for jD_F at any point of E_F . Then $t\bar{a}_w$ satisfies the monic equation whose coefficient of degree $m - k$ is $t^k \bar{c}_k$. These coefficients are regular there. At every point of C_w above this neighborhood, normality therefore makes $t\bar{a}_w$ regular. This proves the required pole bound and hence (7.5); injectivity was established in (7.4). Residue maps themselves respect multiplication.

For the last assertion, work at a selected marked point after splitting. Every $g(a)$ vanishes at every point of its reduced inverse fiber, because this fiber is J -invariant. The non-leading coefficients c_k consequently vanish at the marked point downstairs. There are no poles there, since D_X avoids all the marked points. By (7.3), every $\bar{c}_k$ vanishes at t_i . The reduced monic equation at any point above t_i is then $\bar{a}_w^m = 0$ in its residue field, so $\bar{a}_w$ vanishes there. $\square$

7.2 Residue degrees and the inseparable quotient

For j sufficiently large, the line-bundle Euler-characteristic formula and ample vanishing turn (7.5) into

$$jmn + \chi(Y) \leq jn \sum_{w \in \mathcal{V}} m_w + \sum_{w \in \mathcal{V}} \chi(C_w). \quad (7.7)$$

Here $\deg D_X = \deg D_F = n$, and finite pullback multiplies divisor degree by the map degree. Letting j grow gives $m \leq \sum_w m_w$. On the other hand, Lemma 6.4, applied to the Gauss valuation on $L(X)$ and these distinct prolongations to the degree- m extension $L(Y)$, gives the reverse inequality. Its hypotheses hold: the value group is $\mathbb{Q}$ and the residue field $F(E_F)$ is infinite. We have proved

$$\sum_{w \in \mathcal{V}} m_w = m. \quad (7.8)$$

Let $\Delta_w \leq J$ be the stabilizer of w , and let I_w be the kernel of its action on $K(w)$. The orbit has $m/|\Delta_w|$ members, all of the same residue degree. Therefore

$$m_w = |\Delta_w|. \quad (7.9)$$

In particular this equality follows from the section comparison, before any assertion about the degree of an inseparable quotient.

Lemma 7.4. *The invariant field $M_w = K(w)^{\Delta_w}$ is purely inseparable over $F(E_F)$, of degree*

$$d = [M_w : F(E_F)] = |I_w|. \quad (7.10)$$

In particular d is a power of p and $d \leq m$. If Z_w denotes the normalization of E_F in M_w , the map $C_w \rightarrow Z_w$ is generically separable Galois of degree m_w/d , with group Δ_w/I_w . All these towers are isomorphic over E_F as w varies in $\mathcal{V}$.

Proof. Take $x \in K(w)^{\Delta_w}$. Simultaneous approximation from Lemma 6.4 gives an element $a \in L(Y)$ integral at every member of $\mathcal{V}$, with residue x at w and zero at the other members. The coefficients of the conjugate polynomial $P_a(T)$ in (7.6) have nonnegative Gauss value. At w , its reduction is

$$\overline{P}_a(T) = T^{m-|\Delta_w|}(T-x)^{|\Delta_w|} \in F(E_F)[T]. \quad (7.11)$$

Indeed, elements of Δ_w act on the residue x trivially, and each other conjugate draws its residue from a different member of the orbit, where the prescribed residue is zero.

Write $|\Delta_w| = p^v q$ with $p \nmid q$. In characteristic p ,

$$(T-x)^{|\Delta_w|} = (T^{p^v} - x^{p^v})^q.$$

The coefficient of T^{m-p^v} in (7.11) is therefore $-qx^{p^v}$. The scalar q is nonzero in $F(E_F)$, so $x^{p^v} \in F(E_F)$. This also covers $x = 0$, for which the coefficient is zero. Thus $M_w/F(E_F)$ is purely inseparable.

The effective automorphism group on $K(w)$ is Δ_w/I_w . Finite fixed-field theory gives

$$[K(w) : M_w] = |\Delta_w/I_w|.$$

Combining this with (7.9) and the tower formula proves (7.10). The assertions about the normalizations and their separable function fields follow. Conjugation by J supplies the asserted isomorphisms between the towers, so d is independent of w . $\square$

7.3 Ramification on the residue curves

Pass temporarily to the finite separable splitting field S/F . The curve $Z_{w,S}$ remains integral: its finite map to E_S is surjective and radicial, while separable scalar extension preserves regularity and hence reducedness. The curve $C_{w,S}$ is regular and is a disjoint union of integral components. Each component dominates $Z_{w,S}$: the finite flat map $C_{w,S} \rightarrow Z_{w,S}$ restricts on a component to a nonzero finite locally free summand, of positive rank on the connected target. The components are consequently the normalizations in their respective function fields. These uses of normalization and separable scalar extension are included in Foundations 7 and 8.

Let z_i be the unique point of $Z_{w,S}$ above t_i , and write r_i for its ramification index over t_i . Write a_i for the ramification index of $C_{w,S} \rightarrow Z_{w,S}$ above z_i . We justify that a_i is independent of the point

above z_i , even when $C_{w,S}$ is disconnected. Put $G_w = \Delta_w/I_w$ and $M_{w,S} = S(Z_{w,S})$. The generic algebra of this map is

$$K(w) \otimes_{M_w} M_{w,S} = \prod_{\lambda=1}^t K_\lambda.$$

It is a product of fields because $K(w)/M_w$ is finite separable. The group G_w acts on the first tensor factor, and the invariant algebra is $M_{w,S}$: taking invariants is taking a kernel of linear maps, which commutes with this scalar extension. An orbit of the field factors gives an invariant idempotent. Since $M_{w,S}$ is a field, there can be only one orbit.

The total dimension over $M_{w,S}$ is $|G_w|$, and transitivity makes all factor degrees equal to $|G_w|/t$. The stabilizer of a factor has this same order. It acts faithfully on that factor, since the map from $K(w)$ into the factor is injective. Thus the factor extension is Galois with that stabilizer as its group. Transitivity on primes in each Galois normalization gives transitivity within each component above z_i ; the action on the factors joins these componentwise orbits. We obtain transitivity on the entire fiber of $C_{w,S}$. Ramification indices are preserved by this action. The isomorphisms between the towers for different w give the same r_i, a_i throughout the orbit.

Multiplicities in the tower multiply, so every point P of $C_{w,S}$ above t_i has index $r_i a_i$ over t_i . Finite flatness gives

$$\sum_{P \in C_{w,S}, P|t_i} r_i a_i [\kappa(P) : S] = m_w.$$

Summing over w and using (7.8) yields

$$\sum_{w \in \mathcal{V}} \sum_{P \in C_{w,S}, P|t_i} [\kappa(P) : S] = \frac{m}{r_i a_i}. \quad (7.12)$$

In particular all residue-field degrees, including inseparable ones, are retained in the degree of the reduced fiber.

Apply Lemma 5.4 to $C_{w,S} \rightarrow Z_{w,S}$, component by component. At a point above z_i , the determinant vanishing is at least $a_i - 1 + \mathbf{1}_{p|a_i}$. The total separable degree of the components over $Z_{w,S}$ is m_w/d . The preceding local-degree calculation gives $m_w/(r_i a_i)$ for the sum of the full residue-field degrees above t_i . Hence, for each w ,

$$\nu(C_w) \geq \frac{m_w}{d} \nu(Z_w) + \sum_{i=1}^n \frac{m_w}{r_i a_i} (a_i - 1 + \mathbf{1}_{p|a_i}).$$

Euler characteristics and divisor degrees have the same numerical values over F as after the separable extension to S , so this inequality is written in the original normalization over F . The hypotheses of Theorem 5.1 hold for Z_w/E_F : $[F : F_0]_p < p^h$, and Lemma 7.4 gives a purely inseparable degree $d \leq m$. Divide the displayed inequality by m_w and substitute that theorem's bound for $\nu(Z_w)/d$. Now sum with weights m_w/m , whose sum is one by (7.8). This gives

$$\frac{\sum_{w \in \mathcal{V}} \nu(C_w)}{m} \geq \sum_{i=1}^n \left(1 - \frac{1}{r_i}\right) + \sum_{i=1}^n \frac{a_i - 1 + \mathbf{1}_{p|a_i}}{r_i a_i}. \quad (7.13)$$

Thus the normalization in this inequality agrees with that of (7.2).

7.4 Vanishing conditions force equality

The ramification estimate (7.13) involves the special indices $r_i a_i$, whereas (7.2) involves the generic indices e_i . A second comparison of section spaces will control their difference. Choose the labels in the set

$$\mathcal{I} = \left\{ i : \frac{1}{r_i a_i} > \frac{1}{e_i} \right\}.$$

At these labels, the degree $m/(r_i a_i)$ of the reduced special fiber exceeds the degree m/e_i of the reduced generic fiber. Imposing vanishing there will make the section comparison detect this excess. This selection descends on both sides. The generic index e_i is constant on the label orbits of $\text{Gal}(L'/L)$, and the special indices r_i, a_i are constant on the label orbits of $\text{Gal}(S/F)$. These two permutation actions agree by Lemma 6.7. Hence the same subset of labels defines a divisor over L on the generic base and a divisor over F on the special base.

Let T_Y be the reduced inverse image on Y of this selected divisor, and let T_w be its counterpart on C_w . These are effective divisors on regular curves over their indicated fields. Their degrees may be computed after the finite separable splitting extensions, which preserve reducedness. The generic Galois cover and (7.12) give

$$\deg_L T_Y = \sum_{i \in \mathcal{I}} \frac{m}{e_i}, \quad \sum_{w \in \mathcal{V}} \deg_F T_w = \sum_{i \in \mathcal{I}} \frac{m}{r_i a_i}. \quad (7.14)$$

Restrict the norm to $H^0(Y, \mathcal{O}_Y(jD_Y - T_Y))$. Its integral ball again has a full-dimensional reduction by (7.4), and Lemma 7.3 places that reduction in

$$\bigoplus_{w \in \mathcal{V}} H^0(C_w, \mathcal{O}_{C_w}(jD_w - T_w)).$$

For sufficiently large j , ample vanishing and (7.8) therefore give

$$jmn + \chi(Y) - \deg_L T_Y \leq jmn + \sum_w \chi(C_w) - \sum_w \deg_F T_w.$$

After rearranging and using (7.14), this is

$$\frac{\nu(Y)}{m} \geq \frac{\sum_{w \in \mathcal{V}} \nu(C_w)}{m} + 2 \sum_{i=1}^n \max \left\{ \frac{1}{r_i a_i} - \frac{1}{e_i}, 0 \right\}. \quad (7.15)$$

The three comparisons now force equality. To display their exact interaction, put

$$q_i = \frac{1}{r_i a_i}, \quad b_i = \frac{1}{e_i}, \quad \delta_i = \mathbf{1}_{p|a_i}.$$

The contribution of label i in (7.13) is $1 - q_i + \delta_i q_i$. Combining that inequality with (7.15) and subtracting (7.2) gives

$$\begin{aligned} 0 &\geq \sum_{i=1}^n (b_i - q_i + 2 \max\{q_i - b_i, 0\} + \delta_i q_i) \\ &= \sum_{i=1}^n (|q_i - b_i| + \delta_i q_i) \geq 0. \end{aligned} \quad (7.16)$$

Every summand vanishes, and every intervening inequality is an equality. In particular,

$$r_i a_i = e_i, \quad p \nmid a_i \quad (1 \leq i \leq n), \quad d = |I_w| > 1, \quad \chi(Y) = \sum_{w \in \mathcal{V}} \chi(C_w). \quad (7.17)$$

For the strict inequality $d > 1$, use $p \mid e_i$ and $p \nmid a_i$: then $r_i > 1$, and r_i divides the purely inseparable degree d by finite flatness at t_i . Also $q_i = b_i$ makes the extra term in (7.15) zero, so equality there gives precisely the displayed Euler-characteristic equality.

7.5 A finitely presented model over the valuation ring

Proposition 7.5. *The degree equality (7.8) and the Euler-characteristic equality in (7.17) produce a flat projective scheme $\mathcal{Y}$ of finite presentation over R , with*

$$\mathcal{Y}_L \simeq Y, \quad \mathcal{Y}_F \simeq \coprod_{w \in \mathcal{V}} C_w.$$

Proof. Form the graded R -algebra

$$A = \bigoplus_{j \geq 0} A_j.$$

Multiplication is defined because each w is a valuation: $A_i A_j \subseteq A_{i+j}$. Each piece is finite free by (7.4). Moreover $A_0 = R$, since Y is projective and geometrically integral and hence $H^0(Y, \mathcal{O}_Y) = L$. Put

$$C = \coprod_{w \in \mathcal{V}} C_w, \quad B_j = \bigoplus_{w \in \mathcal{V}} H^0(C_w, \mathcal{O}_{C_w}(jD_w)), \quad B = \bigoplus_{j \geq 0} B_j.$$

The two equalities in the proposition make the source and target dimensions in (7.5) equal for every sufficiently large j . Thus there is an integer j_0 such that

$$A_j / \mathfrak{m}_R A_j \simeq B_j \quad (j \geq j_0), \tag{7.18}$$

compatibly with products. Equality in each high degree, rather than only equality of leading terms, is where the Euler-characteristic equality is used.

We first choose a Veronese regrading for which the reduced algebra is generated in degree one. The divisor on C given by the D_w is ample, so B is finitely generated by [Foundation 9](#). Choose positive-degree homogeneous generators $b_1, \dots, b_t$ over B_0 , of degrees $d_1, \dots, d_t$. Let W be a common multiple of these positive degrees. In a monomial in the b_i , remove pure powers b_i^{W/d_i} , each of weight W , until the remaining exponent of b_i is less than W/d_i . The weight of the remainder is less than tW . Choose $N = qW \geq j_0$ with $q \geq t$.

If the original monomial has degree kN , its remainder has weight cW for an integer $0 \leq c < t \leq q$, since the removed blocks and the original total weight are multiples of W . There are $kq - c$ removed blocks. Combine the remainder with $q - c$ of these blocks to make one factor of weight N , and group the remaining $(k - 1)q$ blocks in groups of q . This factors the monomial into k factors of weight N . Any coefficient from B_0 is absorbed into the first factor. It follows that

$$F \oplus \bigoplus_{k \geq 1} B_{kN} \quad \text{is generated over } F \text{ by } B_N. \tag{7.19}$$

This argument permits $B_0 = H^0(C, \mathcal{O}_C)$ to be larger than F , as can happen for a disconnected curve or a curve with extra constants. All the positive-degree factors used in the argument belong to the full spaces in (7.18).

Let $A^{(N)} = \bigoplus_{k \geq 0} A_{kN}$, with A_{kN} assigned degree k . Choose an R -basis of A_N , and map the variables of a degree-one polynomial ring

$$P = R[x_1, \dots, x_u] \longrightarrow A^{(N)}$$

to that basis. In each degree the target is a finite free R -module. By (7.18) and (7.19), the map is surjective after reduction modulo $\mathfrak{m}_R$. Its degreewise cokernel is finite, so Nakayama's lemma makes the map surjective in every degree. This proves finite generation over R .

For finite presentation, let K_k be the kernel of $P_k \rightarrow A_{kN}$. This surjection splits as an R -module map because A_{kN} is free. Hence K_k is a finite direct summand of the finite free module P_k ,

and reduction gives the exact kernel of the corresponding polynomial presentation over F . The special relation ideal is a homogeneous ideal in the Noetherian ring $F[x_1, \dots, x_u]$, so it has finitely many homogeneous generators. Lift them, in their respective degrees, to elements $g_1, \dots, g_v$ of $\ker(P \rightarrow A^{(N)})$, and let I be the ideal they generate. For every k , the quotient K_k/I_k is a finite R -module. Its reduction is zero, since the reductions of the g_i generate the special relation ideal. Degree-wise Nakayama gives $K_k = I_k$ for every k . Thus

$$A^{(N)} \simeq R[x_1, \dots, x_u]/(g_1, \dots, g_v)$$

is a finite homogeneous presentation. In particular, this argument does not require the nondiscrete valuation ring R to be Noetherian.

Define $\mathcal{Y} = \text{Proj } A^{(N)}$. The degree-one presentation makes it a projective R -scheme of finite presentation. It is flat over R : the graded algebra is a direct sum of free R -modules, homogeneous localization is a filtered colimit of such flat modules, and the degree-zero part of the resulting graded module is a direct summand. Thus every standard affine chart of $\mathcal{Y}$ has a flat coordinate ring over R .

Since A_j spans V_j , the generic fiber is the Proj of the ample section ring of Y after Veronese regrading, and hence is Y . By (7.18), the special fiber is the Proj of the algebra in (7.19). Replacing its degree-zero term F by B_0 has no effect on Proj. Indeed, on a localization at a positive-degree element f , a missing constant $b \in B_0$ is present as $(bf)/f$, since bf has positive degree. The degree-zero coordinate rings of these localizations therefore agree. The full ample section ring recovers C , proving the special-fiber assertion. $\square$

7.6 Connectedness and the final contradiction

The finite presentation just proved is what permits passage to a Noetherian base. This is the finite-equation approximation framework of EGA IV [17, Section 8]; the connected-fiber input is Stein factorization, as in EGA III [16, Section 4.3] and Foundation 13. The following argument constructs the normal Noetherian base explicitly, rather than applying coherent finiteness or Stein factorization to the original valuation ring.

Lemma 7.6. *Let R be a valuation ring of a characteristic-zero field L . If a flat projective R -scheme $\mathcal{U}$ of finite presentation has geometrically integral generic fiber, then its special fiber is geometrically connected.*

Proof. Choose a finite set of homogeneous projective equations for $\mathcal{U}$. Their coefficients generate a finite-type $\mathbb{Z}$ -subdomain $R_0 \subset R$. Normalize R_0 in its fraction field, obtaining R_1 . By Foundation 7, this normalization is finite, so R_1 is a normal Noetherian domain. It remains a subring of R : its elements lie in $\text{Frac}(R_0)$ and are integral over R , which is integrally closed.

The equations define a projective scheme $\mathcal{U}_1$ over R_1 . Its generic fiber is geometrically integral, because this can be checked after extension of its fraction field to L , where the fiber is $\mathcal{U}_L$. Let $\mathcal{T} \subseteq \mathcal{U}_1$ be the scheme-theoretic closure of that generic fiber. Then $\mathcal{T}$ is integral and projective over R_1 : locally its coordinate ring is the image in the coordinate ring of the integral generic fiber, and the generic fiber is dense.

The pullback $\mathcal{T}_R$ is a closed subscheme of $\mathcal{U}$ and agrees with it over L . Its defining ideal in $\mathcal{O}_{\mathcal{U}}$ therefore vanishes upon localization at $R \setminus \{0\}$. On each affine chart, every element of this ideal is annihilated by a nonzero element of R . Flatness of $\mathcal{U}$ makes its affine coordinate rings torsion-free over R , so the ideal is already zero. Thus

$$\mathcal{T}_R = \mathcal{U}. \tag{7.20}$$

Write $K_1 = \text{Frac}(R_1)$ and $\pi : \mathcal{T} \rightarrow \text{Spec } R_1$. Proper coherent finiteness over this Noetherian base gives a finite R_1 -algebra $H^0(\mathcal{T}, \mathcal{O}_{\mathcal{T}})$. Restriction to the generic fiber is injective because $\mathcal{T}$ is integral and that fiber is dense. Since the generic fiber is projective and geometrically integral, its global functions are K_1 . Hence

$$R_1 \subseteq H^0(\mathcal{T}, \mathcal{O}_{\mathcal{T}}) \subseteq K_1.$$

The middle ring is finite over R_1 , so all its elements are integral over R_1 . Normality yields $H^0(\mathcal{T}, \mathcal{O}_{\mathcal{T}}) = R_1$. Equivalently,

$$\pi_* \mathcal{O}_{\mathcal{T}} = \mathcal{O}_{\text{Spec } R_1};$$

one can check this equality on principal affine opens by flat base change for coherent cohomology, as in [Foundation 9](#).

Stein connectedness over the Noetherian base, in the form stated in [Foundation 13](#), now gives geometrically connected fibers of π . The special fiber of $\mathcal{U}$ is, by (7.20), an extension of one of these fibers to the residue field of R . It is therefore geometrically connected. All uses of proper coherent cohomology and Stein factorization in this proof occur over R_1 . $\square$

Completion of the proof of Proposition 7.1. Apply [Lemma 7.6](#) to the flat projective model of [Proposition 7.5](#). Its generic fiber Y is geometrically integral, so its special fiber $\coprod_{w \in \mathcal{V}} C_w$ is connected. Every C_w is nonempty; hence $\mathcal{V}$ has one member. The stabilizer of that member is all of J , so I_w is a normal subgroup of J . By (7.10) and (7.17), it is a nonidentity p -group. This contradicts $O_p(J) = 1$. $\square$

Proof of Theorem 3.1. Fix $h \geq 1$ and choose s_0 satisfying (7.1) with $s = s_0$. The same inequalities hold for every $s \geq s_0$; fix any such s . For any prescribed multiset of conjugacy classes of the p -singular puncture entries, [Lemma 6.5](#) and [Proposition 6.6](#) identify the finite tuple-class set with the corresponding marked-cover classes and supply a Sylow p -group action on that set. If an orbit had size less than p^h , the same proposition would descend one of its covers to a finite extension L/K_s with $[L : K_s]_p < p^h$. This is excluded by [Proposition 7.1](#).

Every orbit of a finite p -group has p -power size. Thus all the orbits have sizes divisible by p^h , and so does their total. The threshold depends only on p, h, m , not on the multiset. An empty tuple set has count zero and satisfies the same assertion; in particular this covers groups with no p -singular elements. $\square$

Proof of Theorem 1.1. The tuple divisibility just proved supplies the hypothesis of [Proposition 3.14](#). Therefore $l = (1 + T)z$ for every pair (X, E) of a finite group and a central block sum considered there. Taking $(X, E) = (G, B)$ gives

$$l(B) = \sum_{\substack{Q \leq G \text{ a } p\text{-subgroup} \\ \text{up to } G\text{-conjugacy}}} z(N_G(Q)/Q, \bar{B}_Q).$$

The normalizer block-assignment calculation following [Lemma 3.2](#) identifies this sum, including $Q = 1$, with $|W_p(B)|$ under the central-character convention (1.1). Hence $l(B) = |W_p(B)|$ for every prime, finite group, and block, as required. $\square$

8 Further consequences

The main theorem gives both numerical information about blocks and a functorial identity. We first prove the local–global bounds stated in [Corollary 1.2](#), then record the more detailed numerical classification and the categorical reformulation. These are consequences of the numerical theorem through the cited results; none is an input to the proof above.

8.1 Local–global and principal-block bounds

The comparison with a defect-group normalizer is local to an individual block. Summing it over maximal-defect blocks gives the comparison with the Sylow normalizer, while the two principal-block bounds relate the number of simple modules to the number of classes of p -elements.

We use the notation of [Corollary 1.2](#); in particular, $P \in \text{Syl}_p(G)$.

Proof of [Corollary 1.2](#). The local inequality and its abelian-defect equality are Alperin’s Consequences 1–2 [1], as restated in [22, Section 3, p. 334]; their numerical weight-conjecture hypotheses follow from [Theorem 1.1](#). Put $N = N_G(P)$. Since P is a normal Sylow p -subgroup of N , it is a defect group of every block of N . Brauer’s first main theorem therefore matches all p -blocks of N with the maximal-defect p -blocks of G ; this indexing is also used in [15, proof of Lemma 2.14]. Writing these two block sets as $\text{Bl}_p(N)$ and $\text{Bl}_{p,\max}(G)$, respectively, the local inequality gives

$$|\text{IBr}_p(G)| \geq \sum_{B' \in \text{Bl}_{p,\max}(G)} l(B') \geq \sum_{b' \in \text{Bl}_p(N)} l(b') = |\text{IBr}_p(N)|.$$

Finally, [Theorem 1.1](#) supplies the only conjectural input in Hung–Sambale–Tiep’s Propositions 3.1–3.2 [22], which give the two principal-block bounds under their respective hypotheses. $\square$

8.2 Character counts and defect data

The next consequence treats blocks whose ordinary-character count exceeds their Brauer-character count by one. Besides giving bounds, it specifies all possible pairs of defect-group order and Brauer-character count, and asserts that every listed pair is attained.

Corollary 8.1 (Character-count and defect data). *Let B be a p -block of a finite group with defect group D of order p^d , and write $k(B) = |\text{Irr}(B)|$. If $k(B) - l(B) = 1$, then the possible numerical data $(p^d, l(B))$ are exactly the following:*

- (i) *There are positive integers n, m such that $d = nm$, every prime divisor of n divides $p^m - 1$, and $4 \mid n$ implies $4 \mid p^m - 1$, with*

$$l(B) = \sum_{e \mid n} \frac{p^{em} - 1}{ne} J_2(n/e), \quad J_2(t) = t^2 \prod_{\substack{q \mid t \\ q \text{ prime}}} (1 - q^{-2}).$$

- (ii) *The exceptional pairs are*

$$(p^d, l(B)) \in \{(5^2, 7), (7^2, 8), (11^2, 9), (11^2, 35), (23^2, 88), (29^2, 63), (59^2, 261)\}.$$

All listed data occur, and in every case $k(B) = l(B) + 1$.

Proof. Hung, Sambale, and Tiep proved this numerical classification assuming the numerical weight conjecture for B [22, Theorem 3.3]. [Theorem 1.1](#) supplies that hypothesis. Their examples establishing that every listed value occurs were already unconditional. $\square$

This classifies only character-count and defect numerical data, not blocks up to isomorphism, Morita equivalence, or derived equivalence.

8.3 Functorial formulation

The following formulation expresses the numerical theorem as an identity of classes of functors. Its alternating sum runs over all strict chains; it is not the normal-chain transform used earlier to prove the theorem.

The numerical theorem gives the functorial formulation of Boltje, Bouc, and Yilmaz [4, Conjecture 3.3]. Let k and F be algebraically closed fields of characteristics p and zero, respectively. We recall the objects used in this optional formulation. A finite-dimensional (kH, kG) -bimodule is viewed as a $k[H \times G]$ -module by $(h, g)m = hmg^{-1}$. It is a p -permutation module if its restriction to a Sylow p -subgroup has a basis permuted by that subgroup. Such a p -permutation bimodule is diagonal if it is also projective as a left kH -module and as a right kG -module. The notation $T^\Delta(H, G)$ denotes their split Grothendieck group, with relations $[M \oplus N] = [M] + [N]$.

The category of group–central-idempotent pairs has objects (H, c) , including $c = 0$. A morphism from (H, c) to (H', c') is an F -linear combination of diagonal p -permutation bimodule classes selected by c' on the left and c on the right. Composition is induced by tensor product over the intermediate group algebra. In the notation of [4, Section 2], the category of F -linear functors from this category to F -vector spaces is $\mathcal{FB}\ell_{F,k}$.

This functor category is semisimple in characteristic zero [4, Section 2.5]. By Yoneda, the endomorphism space of the representable $FT^\Delta(-, H)$ is the finite-dimensional F -space $FT^\Delta(H, H)$. Semisimplicity therefore forces the representable to have finite length: each simple summand contributes an independent projection. Its direct summands also have finite length. Here $K_0(\mathcal{FB}\ell_{F,k})$ denotes the Grothendieck group of the finite-length subcategory, the free abelian group on simple-functor classes. It is in this sense that classes record the simple multiplicities used in [4, Sections 2.5 and 3].

The bimodule kHc acts by precomposition on the representable functor $FT^\Delta(-, H)$. Its image is the direct summand $FT_{H,c}^\Delta = FT^\Delta(-, H) \circ kHc$ associated to (H, c) , using the equivalent group and pair descriptions in [4, Sections 2.3–2.4]. In particular $c = 0$ gives the zero functor. Write $[[H, c]]_F$ for its class in $K_0(\mathcal{FB}\ell_{F,k})$.

Write $S_{1,1,F}$ for the simple functor indexed by the trivial pair and its trivial F -module. Its multiplicity in $FT_{H,c}^\Delta$ is $l(H, c)$: evaluation at the trivial group gives the space spanned by indecomposable projective kHc -modules [4, proof of Theorem 3.4]. Thus the following identity records all simple-functor multiplicities, with the simple-module count as its $S_{1,1,F}$ component.

Corollary 8.2 (Functorial Alperin weight identity). *Let b be a block idempotent of kG , where G is finite. Let $\mathcal{S}_p(G)$ consist of all strict chains $\sigma = (1 = P_0 < P_1 < \cdots < P_n)$ of p -subgroups, including (1). Put*

$$|\sigma| = n, \quad G_\sigma = \bigcap_{i=0}^n N_G(P_i), \quad b_\sigma = \text{Br}_{P_n}(b).$$

Since $C_G(P_n) \leq G_\sigma$ and the Brauer image is invariant under G_σ , the element b_σ is a central idempotent of kG_σ , possibly zero or a sum of blocks. Thus the pair (G_σ, b_σ) is an object of the category just described. Then, summing over representatives of the G -conjugacy classes of chains,

$$\sum_{\sigma \in [G \backslash \mathcal{S}_p(G)]} (-1)^{|\sigma|} [[G_\sigma, b_\sigma]]_F = \begin{cases} [S_{1,1,F}], & \text{if } b \text{ has defect zero,} \\ 0, & \text{if } b \text{ has positive defect} \end{cases} \quad \text{in } K_0(\mathcal{FB}\ell_{F,k}).$$

Proof. Boltje–Bouc–Yilmaz first prove that, for every group–block pair, the alternating sum above is an integer multiple of $[S_{1,1,F}]$: every other simple-functor multiplicity cancels [4, Theorem 3.5(1)]. The universal numerical blockwise Alperin weight conjecture determines the remaining coefficient

through its chain formulation. Their Theorem 3.5(2) therefore gives the stated identity from Theorem 1.1. $\square$

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